



Aircraft maintenance scheduling optimization: state-of-the-art and emerging trends

Zimu Guo, Xin Wen^{*}, Xinyu Huang

Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Hong Kong Special Administrative Region

ARTICLE INFO

Keywords:

Aircraft maintenance scheduling
Aviation operations research
Optimization frameworks
State-of-the-art review

ABSTRACT

Aircraft maintenance is essential to ensuring airworthiness, service reliability and flight punctuality in the aviation industry, which are critical to aviation safety, operation continuity and passenger satisfaction. To achieve these objectives efficiently, aircraft maintenance scheduling (AMS) plays a crucial role in coordinating maintenance activities with complex flight schedules by optimizing resource allocation, reducing downtime, and operational cost. Despite the importance of AMS, comprehensive review of AMS research remains limited, particularly with respect to the evolution and application of advanced scheduling approaches. To address this gap, this study systematically reviews mathematical-model-based approaches published in leading operations research and transportation journals between January 2014 and December 2025. This review synthesized a wide range of literature and organized it into three key areas: maintenance decision-making, solving strategic and tactical choices in task priority and resource allocation; Optimization model, including deterministic, stochastic and dynamic methods to balance efficiency, cost and robustness; And solutions, including exact algorithms, heuristics and data-driven methods. This structural analysis provides clear insights for scheduling decisions and emphasizes the progress of modeling technology and its practical significance. This paper also identifies key research gaps and suggests future research directions for the AMS. By integrating different solution approaches and key findings, this review provides a comprehensive theoretical and operational overview of the current state of the art in AMS.

1. Introduction

Aircraft has become one of the primary transportation modes for both passengers and freight worldwide (Ben Ahmed et al., 2017; Cadarso & Vaze, 2023). Reliable operating conditions are critical for aircraft, and maintenance serves as a vital role in ensuring aviation safety, flight punctuality, and cost-effectiveness for airlines (Van Kessel et al., 2023). In 2023, global airlines spent approximately \$93.9 billion on maintenance operations, underscoring the immense scale and economic significance in this domain.¹ Inefficient maintenance scheduling can lead to flight delays and disruptions, posing persistent challenges for airlines (Lagos et al.,

^{*} Corresponding author.

E-mail addresses: ethan.guo@connect.polyu.hk (Z. Guo), windy.wen@polyu.edu.hk (X. Wen), xinyu-97.huang@connect.polyu.hk (X. Huang).

¹ International Air Transport Association (IATA), "Airline Maintenance Cost Executive Commentary: FY2023 Data," <https://www.iata.org/contentassets/8437020db31a4717b70677d9b06b1a45/fy2023-mcx-report-public.pdf>.

2020). The annual global economic loss due to unscheduled aircraft downtime is estimated to exceed \$50 billion,² highlighting the critical role and substantial improvement potential of effective maintenance management in airline operations. These substantial economic and operational impacts emphasize the need for maintenance stakeholders to optimize decisions, not only for cost reduction but also for enhancing operational efficiency, safety standards, and customer satisfaction. As the global aviation faces expanding fleet sizes, increasing traffic demand, and more stringent regulatory requirements, optimizing AMS has become essential for ensuring aviation safety, aircraft reliability, and cost efficiency (Eltoukhy et al., 2022).

Maintenance operations are among the most critical tasks in aviation, with associated challenges stemming from multiple factors. AMS defines appropriate maintenance opportunity to aircraft with corresponding resources to ensure airworthiness while balancing operational efficiency and economic costs. Airlines are obligated to meet stringent safety regulations which are established by regulatory entities such as the Federal Aviation Administration (FAA) under 14 CFR³ and the European Union Aviation Safety Agency (EASA) under Commission Regulation (EU) 2018/1139.⁴ These entities regulate inspection and repair operations across different maintenance strategies, including preventive, corrective, and predictive approaches. Preventive maintenance (PM) schedules maintenance proactively according to time intervals or usage conditions, while corrective maintenance (CM) deals with unexpected breakdowns (Gustavsson et al., 2014; Jordan et al., 2022). Predictive maintenance (PdM) leverages advanced data analytics and AI-based techniques to forecast equipment health, optimize maintenance operations and reduce unnecessary downtime.

AMS amplifies these complexities when incorporating with various airline planning problems, as shown in Fig. 4, which create complex interdependencies that challenge traditional scheduling approaches and complicate planning (Shao et al., 2017; Cadarso & Vaze, 2023). On the one hand, the airlines prioritize aircraft availability and flight punctuality, integrating flight and maintenance schedules to ensure consistency and regulatory compliance. On the other hand, Maintenance, Repair, and Overhaul (MRO) providers emphasize efficient resource utilization and service punctuality to strike a balance between profitability and customer satisfaction. Across various planning horizons, aircraft maintenance spans strategic, tactical, and operational decision-making levels. Decisions across these time periods further complicate planning efforts.

While recent literature has begun to address specific sub-problems, such as the review by Ma et al. (2022) which focuses on uncertainties in Aircraft Maintenance Routing (AMR), there remain a notable lack of comprehensive survey that summarizes the broader state-of-the-art in AMS optimization. Inspired by this shortfall and the escalating relevance of the subject, we offer a systematic literature review. This paper examines the cutting-edge literature within the field of aircraft maintenance scheduling, encompassing strategies, maintenance stakeholders, maintenance categories, various decision levels and domains. It examines deterministic, stochastic, and dynamic models, assessing their ability to address uncertainties with various constraints and performance measures. This review assists both researchers and practitioners in understanding the foundations of aircraft maintenance planning. By identifying research gaps in current studies, this study seeks to encourage academia and industry to put forth greater effort and contribute to this crucial research area.

The remainder of this review is structured as follows. Section 2 introduces all details on the research methodology. In section 3, maintenance decision-making is discussed from different perspective, including strategies, maintenance stakeholders, and maintenance categories. Additionally, decision levels across different planning horizons and domains related to integrated problems are explored. Section 4 illustrate emerging optimization models and solution approaches in AMS. In Section 5, we outline further discussion, and conclusions are drawn regarding aircraft-maintenance-related issues.

2. Methodology

Motivated by the critical importance of AMS, while other reviews primarily emphasize broader perspectives, we focus on research papers employing operations research and data-driven approaches to address maintenance scheduling challenges. These papers include deterministic and stochastic formulation, with most models expressed as mixed-integer programming (MIP) and integer programming (IP). Following the logic presented in the abstract and introduction, this review examines aircraft maintenance scheduling (AMS) research from a problem-to-solution perspective. To support this analysis, key modelling characteristics and algorithmic methods are systematically summarized in Tables 3 and 4 of appendix. Based on this structure, the literature search and classification are designed to support a clear and consistent review framework.

This review adopts search and selection methodologies similar to those described in prior studies, emphasizing premier publication journals in operations research and management science, as shown in Table 1 (appendix). Keyword searches were conducted in different databases, targeting the title, keywords, and abstract fields with the terms (“aircraft maintenance”) AND (“schedule” OR “plan”). To align with our objectives, we restrict our review to publications that focused on optimization and data-driven approaches to AMS. Overall, this review covers and discusses a total of 104 papers.

As illustrated in Fig. 1, the number of publications in the AMS field has remained consistently active from January 2014 to December 2025. Researchers’ interests in this field have shown growth since 2018. Notably, 20 papers were published during 2018–2019, indicating the growing importance and research momentum of AMS.

Table 2 in the appendix provides summary data for the collected articles. Among these, Computers & Industrial Engineering

² AAA Air Support. "Minimizing Downtime: How AOG Events Impact Airlines Like Delta." <https://www.aaaairsupport.com/minimizing-downtime-how-aog-events-impact-airlines-like-delta/>.

³ Federal Aviation Administration, 14 CFR Part 145 — Repair Stations, [drs.faa.gov](https://www.faa.gov).

⁴ European Union Aviation Safety Agency, Commission Regulation (EU) No 1139/2018 — <https://www.easa.europa.eu/en/regulations>.

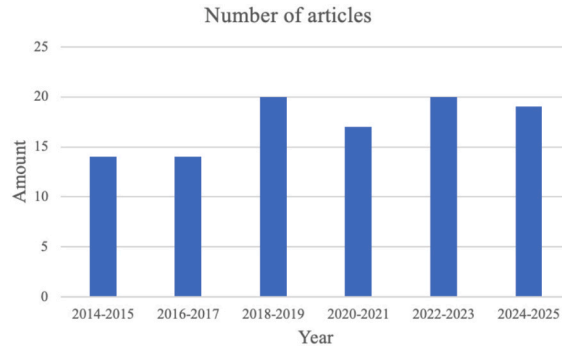


Fig. 1. Number of articles published from January 2014 to December 2025.

emerges as the leading contributor in the aircraft maintenance domain, accounting for 21 articles. European Journal of Operational Research rank closely behind, followed by Journal of Air Transport Management, Computers & Operations Research and Transportation Science. Collectively, these journals account for over half of the analyzed articles. Additional papers optimizing AMS are predominantly featured in outlets such as International Journal of Production Research, Applied Soft Computing, Transportation Research Part C: Emerging Technologies, Annals of Operations Research and Decision Support Systems. Each publication falls within the scope of OR/OM.

We selected recently published articles from mainstream OR/OM/MS journals for several reasons. First, these journals are highly reputable, ensure timely coverage of recent advances and are firmly grounded in the relevant research field. Second, by prioritizing mainstream journals, we ensure comprehensive coverage of research outlets. Third, the distribution of publications by year reveals significant research opportunities across various topics within the aircraft maintenance domain. Fourth, the search process identifies both widely explored and underexplored topics, as well as recent advancements in methodologies and technologies, such as optimization algorithms and machine learning, providing directions for future studies.

Based on the reviewed literature, we organized the AMS studies into two main themes: (a) The first theme focuses on the maintenance decision-making in AMS, including maintenance strategies, stakeholders, maintenance categories, decision levels with varying planning horizons, decision domains in both maintenance operations and integrated planning problems. (b) The second theme concerns modeling and solution aspects of AMS, covering objective functions and constraints, and solution approaches. Our goal is to present these classification themes clearly and accessibly, thereby enabling readers to understand the specific concerns of different stakeholders in AMS. The first theme clarifies issues related to scheduling decisions and encompasses six distinct characteristics. These operational characteristics are essential for understanding and addressing the core elements discussed in the reviewed articles. In the second theme, we synthesize the modeling methodologies and solution approaches employed in the reviewed papers.

3. Maintenance decision making in aircraft maintenance scheduling

3.1. Maintenance strategies

The aviation maintenance field relies on a series of strategic methods, including CM, PM, and PdM, to maintain flight safety, ensure system reliability, and optimize resource management in the increasingly complex aviation maintenance configuration. This section explores these core maintenance strategies, which serve as the fundamental pillars of effective maintenance planning and execution in the aviation sector. CM is a reactive strategy performed after a functional failure or fault detection, aimed at restoring system functionality. CM presents significant challenges for resource allocation due to its inherent unpredictability. In contrast, PM presents a proactive, scheduled approach where tasks are conducted according to fixed intervals, such as flight hours, cycles or calendar days, in order to prevent deterioration and maintain system availability, laying the foundation for high reliability mechanisms in modern aviation systems. Finally, PdM leverages multi-source real-time data and advanced modeling to forecast the RUL of components. Unlike the static thresholds of PM, PdM enables dynamic scheduling by predicting failures before they occur, providing more efficient and resilient operational planning.

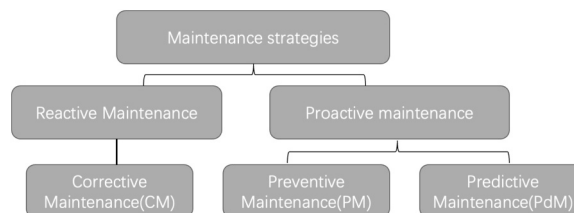


Fig. 2. Overview of maintenance strategies.

3.1.1. Corrective maintenance

CM in the aviation maintenance system refers to a range of maintenance activities that are executed following the malfunction or failure to restore the system or equipment to its functional state. As a critical component for ensuring flight safety and operational continuity, CM is widely used in line maintenance, base maintenance, and ground support operations. It is primarily applied to address unplanned failures, emergencies, or operational disruptions. Unlike scheduled PM or prediction-driven maintenance strategies, CM is a reactive approach typically triggered passively, shown in Fig. 2, with concentrated resource demands, and has a significant disruptive impact on operational rhythms, particularly when maintenance resources are constrained, or system reliability assessments are insufficient. Under such conditions, the frequency of CM occurrences tends to increase significantly. Although CM is indispensable for restoring aircraft to an operational state, its sudden nature and inherent uncertainty pose considerable challenges to maintenance planning, flight stability, as well as inventory and manpower allocation. Consequently, research on CM has become a key focus of extensive scholarly attention and exploration.

In practical applications, CM is a prominent feature in the operations of airlines, fulfilling emergency recovery functions. Jordan et al. (2022), in their study of KC-135R tanker aircraft availability forecasting, noted that CM activities, as typical unplanned maintenance events, lead to the redistribution of maintenance resources, adjustment of task priorities, and increased burden on the component supply chain. These effects are indirectly reflected in operational metrics such as break rate, AWM delay defect waiting rate, and 12-hour repair rate. Similarly, Van Kessel et al., (2023), in their study on flight maintenance task rescheduling, incorporated CM tasks into a rolling rescheduling mechanism and found that frequent CM events were a primary factor in disrupting the stability of planned maintenance schedules. Therefore, how to model, schedule, and strategically mitigate CM behaviors is a critical issue in maintenance optimization.

Research on CM strategies primarily focuses on its behavioral modeling and execution optimization. On one hand, Sharma & Rai (2021) modeled CM as an imperfect maintenance process, where the post-maintenance state of the equipment may not fully restore to as good as new but rather is subject to a quality limitation in the repair process, thus affecting the future reliability evolution. The paper introduces a CM repair quality coefficient q_{CM} based on the Kijima virtual age model and constructs a maximum likelihood function to estimate model parameters under censored data conditions, reflecting the dynamic role of CM in the system's degradation path. On the other hand, the triggering mechanism of CM and its response costs are systematically incorporated into maintenance strategy optimization models. Regattieri et al. (2015) introduced CM as a high-cost, uncertain factor in their proposed maintenance strategy evaluation framework, providing a quantitative reference for maintenance decision-making by systematically calculating the CM frequency and repair costs for each component. Additionally, with the development of intelligent maintenance techniques, CM has also been included in the comparative scope of data-driven strategy optimization. Hu et al. (2021) incorporated CM as one of the foundational actions in a reinforcement learning model, treating it as a passive maintenance strategy and comparing it with scheduled maintenance, PHM-based maintenance strategies, and reinforcement learning strategies across multiple environments. Their experiments show that the CM strategy is inferior in task completion rate, cost control, and adaptability, highlighting that although CM is unavoidable, its proportion should be controlled through strategic optimization to improve overall system performance.

In summary, CM plays an indispensable yet cautious role in aviation maintenance. It is not only a direct response to unforeseen failures but also a key variable in assessing system availability risks and resource scheduling flexibility.

3.1.2. Preventive maintenance

PM is a strategy for performing planned maintenance on equipment before it has experienced a manifest failure, based on a predetermined time, intensity of use, duty cycle, or reliability threshold. It's designed to delay deterioration, reduce the risk of unexpected failures, and enable maintenance resources allocation while safeguarding system availability (Levi et al., 2014). Different from CM that only responds after a failure, PM emphasizes proactivity and plannability, and provides a foundation for high-reliability maintenance operations in modern aviation systems.

In practice, PM is usually scheduled based on two types of metrics: first, time-based metrics, such as cumulative flight hours or months, the number of flight cycles, and mission cycles, which correspond to the basis of cyclic overhaul strategies; and second, condition and utilization-based metrics, such as Mean Time Between Failures (MTBF), component deterioration rate, historical maintenance intervals, and health status assessment results, which are used to support more granular schedule adjustments (Sharma & Rai, 2021; Jordan et al., 2022). In practice, the effective implementation of PM often requires a combination of operational requirements such as task package requirements, maintenance window length, base capacity, and aircraft spare parts availability (Seif & Yu, 2018; Van Kessel et al., 2023).

Disturbance factors in aviation operations, such as flight delays and flight timetable changes, can have an impact on the scheduled execution of PM tasks. For this reason, some studies have proposed embedding PM tasks in a rolling optimization and rescheduling framework to enhance the robustness and execution rate of maintenance schedules (Van Kessel et al., 2023). Meanwhile, the economics of PM strategies have also received much attention. For example, Gustavsson et al. (2014) established a joint modeling of setup-related cost and interval-based cost in the model, revealing the possibility of cost minimization through coordinated multi-component maintenance, while Golbasi & Turan (2020) further verified through discrete-event simulation that the rational scheduling of PM tasks under resource restriction scenarios can significantly improve system availability and reduce the risk of default.

In scenarios with highly complex or modularized system structures, the scheduling of PM strategies needs to be further combined with the disassembly and assembly dependencies among components. Levi et al. (2014) developed a maintenance scheduling framework relying on a submodular cost function for coordinating the maintenance frequency of different modules and resolving conflicts and redundancies among system hierarchies, while Regattieri et al. (2015) constructed a three-phase maintenance optimization framework based on actual A320 fleet operation data, taking into account reliability, maintenance simulation and inventory

configuration, and verified that the annual cost can be substantially reduced to a fractional of the original plan through refined PM strategy scheduling, while guaranteeing an outstanding level of the system availability.

Given the escalating complexity within aviation maintenance configurations, the application of PM has gradually evolved into a multi-dimensional decision-making progress that integrates structural modeling, operation scheduling and resource coordination. Its implementation not only relies on static rules and empirical thresholds, but also requires multi-objective optimization by combining real-time status, post-maintenance impacts and task priorities.

3.1.3. Predictive maintenance

With the increasing demand for intelligent maintenance and resource optimization management in high-reliability systems such as aviation, PdM has gradually become a key focus in maintenance strategy research. Based on existing studies, PdM is commonly defined as a strategy that leverage multi-source operational data and data-driven modeling techniques to dynamically predict equipment health status and potential failures, thereby enabling timely maintenance interventions. Its essence lies in proactively planning maintenance windows and resource allocation by anticipating failure signs or health degradation trends in advance, minimizing system disruption risks and maintenance costs (Aremu et al., 2020; Wang et al., 2024).

Compared to PM, which is based on fixed time intervals or usage cycles, PdM emphasizes data-driven and condition-based monitoring, adapting more finely to the actual degradation process of equipment. In practical applications, its core advantages include reducing unnecessary maintenance, avoiding conflicts in maintenance resource allocation, improving the efficiency of manpower and spare parts configuration, and extending the effective service life of equipment. For example, Wang et al. (2024) proposed embedding the Remaining Useful Life (RUL) results predicted by deep learning into a multi-objective scheduling model, effectively reducing maintenance conflicts and task delays. Similarly, the ForeSim-BI system developed by (Dinis et al., 2020) integrated a prediction module with a Bayesian network to achieve dynamic matching of maintenance capacity and resource demands, enhancing the responsiveness and accuracy of maintenance planning.

At the strategy level, PdM encompasses various modeling paths and decision-making mechanisms. Some studies use time series modeling techniques (e.g., auto-regressive moving average (ARMA) models) to predict occurrence time of failure events and optimize them with machine learning algorithms. For instance, Baptista et al. (2018) integrated ARMA modeling with Support Vector Regression (SVR) for short-term fault prediction of aircraft engines. Other scholars have developed feature selection and health assessment systems to enhance the generalization ability and robustness of models (Aremu et al., 2020; Sharma & Rai, 2021), or designed unsupervised learning-based algorithms to continuously learn and update state data representations (Savitha et al., 2020). To tackle data class imbalance in real-world datasets, Dangut et al. (2022) formulated balanced calibration ensemble method which significantly improved the minority-class fault detection.

Moreover, for non-interruptible maintenance tasks, combing PdM strategies with optimal age replacement models leads to more cost-effective maintenance planning.. In the context of task-oriented systems, Zhang et al. (2023) constructed an age replacement model with the help of a discount factor and proposed a strategy framework for proactive replacement based on task gaps, balancing long-term economic benefits and task continuity, thus expanding the adaptability of PdM strategies in special operational environments.

In summary, PdM, as a modern maintenance strategy integrating data sensing, intelligent prediction, and decision optimization, not only demonstrates significant advantages in improving system availability and reliability but also provides a foundation for more resilient and cost-benefit-driven maintenance models. The strategic methodology for PdM is continuously expanding, covering various aspects such as fault prediction, life assessment, anomaly detection, and resource scheduling, providing a solid foundation for subsequent comparative and integration research with other maintenance models.

3.2. Maintenance stakeholders

AMS is a complex process driven by the interaction of different stakeholders, primarily airlines and MRO providers, which have distinct interests and goals. Acting as the main operator of aircraft, airlines prioritize flight safety, operational reliability, availability, and punctuality, seeking to minimize maintenance costs and flight schedule disruptions. In contrast, MRO providers focus on optimizing resource utilization, ensuring compliance with service level agreements among various airlines, and striking a balance between cost efficiency and service quality. These different priorities often lead to conflicts in task prioritization, resource allocation, and risk management, requiring complex coordination mechanisms.

3.2.1. Airlines

As the primary stakeholder of aircraft maintenance, airlines prioritize maintenance decisions that maximize aircraft availability and ensure punctual operations in order to meet operational requirements and enhance customer satisfaction. In addition, airlines play strong emphasis on minimizing maintenance costs and tend to optimize overall scheduling, real-time adjustment of maintenance plans, and resource allocation to ensure prompt response to breakdowns, thereby reducing downtime losses and maintaining operational efficiency (Shaikat et al., 2020). To achieve this goal, airlines must strike a precise balance between maintenance operations and flight scheduling by carefully planning the execution windows of maintenance tasks to reduce aircraft downtime and maintenance disruptions to flights (Cacchiani & Salazar-González, 2017; Faust et al., 2017). At the same time, airlines have actively pursued improvements in maintenance operation through rational allocation of maintenance resources, optimal inventory management, and effective personnel allocation.

In selecting decision-making models and methods, airlines increasingly adopt holistic approaches that integrate flight scheduling

with maintenance resource allocation to coordinate different operational activities. For example, airlines employ approximate dynamic planning strategies to manage the real-time dynamics of maintenance operations and the uncertainty of resource requirements (Lagos et al., 2020). Furthermore, recent trends show a shift toward passenger-centric scheduling, where maintenance windows are strategically embedded into the initial flight design process to enhance overall system performance (Faust et al., 2017).

In operation practice, conflicting interests between different stakeholders create a strong need for coordination, particularly in task prioritization and resource allocation strategies. Airlines usually prioritize high flexibility and rapid response capabilities in maintenance resources to ensure operational continuity and timeliness; while maintenance service providers tend to maintain the stability of maintenance tasks and balanced allocation of resource load (Qin & Ng, 2023). To address this conflict, recent studies have proposed various approaches to promote win-win cooperation between the two parties in maintenance resource allocation and task scheduling, including collaborative mechanisms, information-sharing platforms and coordination models based on game theory (Qin & Ng, 2023).

In addition, airlines maintain a more conservatism attitude towards operational risks than maintenance service providers, which prompts them place a greater emphasis on the development of maintenance task scheduling. Airlines introduce conditional value-at-risk (CVaR) mechanisms and front-loaded buffer management strategies to guard against operational risks arising from demand fluctuations and unexpected events (Liu et al., 2019; Öhman et al., 2021). Furthermore, airlines also actively adopt strategies such as rolling planning, real-time rescheduling and contingency resource in scheduling and maintenance tasks to ensure the flexibility and robustness of their maintenance operations (Lagos et al., 2020).

3.2.2. MRO providers

Maintenance, Repair, and Overhaul (MRO) providers, serving as key stakeholders within the aircraft maintenance domain, primarily focus on maximizing resource utilization efficiency, enhancing profitability through cost control, and ensuring compliance with service level agreements (SLAs). To achieve an efficient resource allocation plan, MRO providers typically develop optimization formulations to coordinate limited resources such as hangars, technicians, and spare parts, thereby reducing customer waiting times, maintenance lead times, and enhancing service punctuality (Ertogral & Öztürk, 2019; Qin et al., 2020). Furthermore, when addressing simultaneous maintenance demands from multiple airlines, MRO providers leverage multi-objective formulations or multi-client priority decision models, employing refined resource allocation and task sequencing to meet diverse demands effectively. In practice, heuristic algorithms and data-driven approaches further provide computationally efficient alternatives for solving maintenance scheduling problems, facilitating timely decision-making (Kurz, 2016; Qin et al., 2018; Dinis et al., 2020).

In applying specific decision-making methods, MRO providers adopt various optimization and decision-support tools tailored to different operational scenarios to achieve rational resource allocation and task scheduling. For instance, Ertogral & Öztürk (2019) proposed an integrated decision framework focusing on inventory management and workforce allocation to reduce maintenance lead times and operational costs. Dinis et al. (2020) developed a decision-support system by integrating maintenance task specifications and resource assessment methods to improve resource utilization efficiency and service responsiveness. Additionally, Qin et al. (2018) addressed complex scenarios during maintenance peak periods, proposing optimization strategies that balance economic benefits with safety requirements, demonstrating the applicability and effectiveness of decision tools in operational contexts.

To balance the tradeoff between economic interests and customer satisfaction, MRO providers must mitigate conflicts effectively within maintenance contracts and establish robust coordination mechanisms. Under performance-based contracts, conflicts regarding maintenance cycles, inventory levels, and permissions for early maintenance frequently arise between providers and clients. MRO providers must negotiate reasonably to mitigate the negative impacts of premature maintenance on clients' operational cycles (Ertogral & Öztürk, 2019). Qin & Ng (2023) further noted that implementing priority decision mechanisms and collaborative models between airlines and MRO providers significantly enhances resource allocation flexibility and service quality, alleviating conflicts arising from resource requirements and fostering mutually beneficial outcomes for stakeholders. To capture the characteristics in maintenance outsourcing, recent studies have modeled the interaction between airlines and MRO providers using bilevel optimization (Sun et al., 2025). They proposed a bilevel decision-making model for maintenance outsourcing cooperation, in which airlines act as upper-level decision makers determining outsourcing strategies and service requirements, while MRO providers optimize resource allocation and operational decisions at the lower level. Their framework effectively coordinates conflicting objectives between cost control and service quality, providing a structured mechanism to improve cooperation efficiency and contract performance in maintenance outsourcing arrangements.

Furthermore, risk management is a critical challenge for MRO providers in long-term operations, encompassing supply chain stability, workforce availability, and equipment reliability. Singh et al. (2024) integrated multi-agent simulation and soft systems methodology (SSM) to identify critical risk sources, such as no-fault-found (NFF) events, in the maintenance process, emphasizing the necessity of the integration of risk cost assessments into contractual framework to enhance the resilience of long-term contract execution. From a regulatory perspective, Nam et al. (2023) introduced a safety compliance framework covering a couple of dimensions, including technicians, tools, equipment, and incident investigation, to promote standardized operations and improve MRO providers' risk response capabilities. Kurz (2016) emphasized that incorporating advanced prediction information, such as real-time equipment status and maintenance demand, can significantly improve maintenance prediction accuracy and responsiveness, thereby reducing resource bottlenecks and uncertainty risks during maintenance processes.

In summary, as critical stakeholders in the maintenance service ecosystem, MRO providers must comprehensively address resource allocation efficiency, cost control, risk management, and compliance requirements. Through precise decision models and coordination mechanisms, an optimal balance between their own interests and customer satisfaction can be attained, thereby ensuring long-term operational stability and profit maximization (Kurz, 2016; Qin et al., 2020).

3.3. Maintenance categories

Regular inspections ensure flight safety, system reliability, and operational efficiency through various maintenance categories, with Line and Base maintenance constituting the two primary components. Line maintenance focuses on short-term, lightweight tasks of aircraft during flight intervals, aimed at maintaining airworthiness and operational continuity. It is typically carried out at airports or hub stations and has high time sensitivity and scheduling complexity. In contrast, base maintenance involves large-scale, systematic tasks that require downtime, which need to be carried out in maintenance bases equipped with professional facilities, emphasizing resource integration and long-term planning. A simple illustration is demonstrated in Fig. 3.

3.3.1. Line maintenance

Line Maintenance involves lightweight maintenance tasks performed during brief aircraft turnaround periods between flight segments, including routine inspections, fluid replenishment, initial fault diagnosis and rectification, component replacement, and pre-flight/post-flight checks. Such tasks are characterized by high frequency, constrained time windows, and minimal reliance on large-scale facilities, typically conducted at airline operating bases, hub airports, or major transit stations (Shaukat et al., 2020; Öhman et al., 2021). These operations are primarily executed by airline in-house maintenance teams or third-party MRO providers stationed at airports, emphasizing flexible scheduling of multi-skilled technician and efficient coordination of on-site resources (Sciau et al., 2024). Line maintenance tasks must be completed within the stringent timeframes imposed by flight schedules, exhibiting scheduling complexity and significant time sensitivity, often necessitating real-time integration with flight planning systems to mitigate delays and resource conflicts (Lagos et al., 2020).

Advancements in modern aircraft system reliability and maintainability have facilitated the delegation of certain repetitive tasks, traditionally performed during periodic A-checks (see Section 3.3.2 for details) within base maintenance, to line maintenance. This shift alleviates the burden on base maintenance operations, enhances maintenance planning flexibility, and improves aircraft utilization efficiency (Gopalan, 2014; Öhman et al., 2021). Consequently, line maintenance has evolved from a reactive, interim remedial mechanism into a critical maintenance component requiring systematic planning and optimization. Recent studies have proposed various methodologies (Lagos et al., 2020) to support airlines in rationally allocating maintenance resources, configuring task buffers, and completing task timely under operational pressures. Furthermore, research on aviation maintenance network design underscores the importance of optimizing the spatial distribution of line maintenance stations in alignment with route networks to ensure the seamless execution of routine maintenance tasks during aircraft turnarounds (Gopalan, 2014).

3.3.2. Base maintenance

In both industrial and scholarly context of aviation maintenance, the terms base maintenance, heavy maintenance, and hangar maintenance are often used interchangeably to describe planned maintenance tasks requiring aircraft grounding, component disassembly, in-depth system inspections and repairs. Although each term emphasizes distinct aspects—base maintenance highlights execution at designated maintenance bases (Liu et al., 2019), Hangar maintenance underscores the facility context (Van Kessel et al., 2023), and heavy maintenance emphasizes task complexity—these terms collectively refer to the same category of intensive maintenance activities. Consequently, these terms are typically treated as a group highly overlapping concepts covering the comprehensive maintenance tasks associated with C-checks and D-checks.

Based on the standardized maintenance classification system, aircraft maintenance activities are typically categorized into four different check types (A, B, C, and D) according to their complexity and cycle limits. A-checks, the most frequent routine inspections, are conducted approximately every 500 to 800 flight hours or 60 to 100 flight days, involving visual inspections, fluid replenishment, and filter replacements. Some of these tasks can be performed during flight turnarounds or overnight stops. Recent studies note that these tasks are increasingly integrated into line maintenance. B-checks, historically considered an intermediate tier between A-checks and C-checks, were typically scheduled every 4 to 6 months and involved slightly more in-depth tasks, such as functional tests or

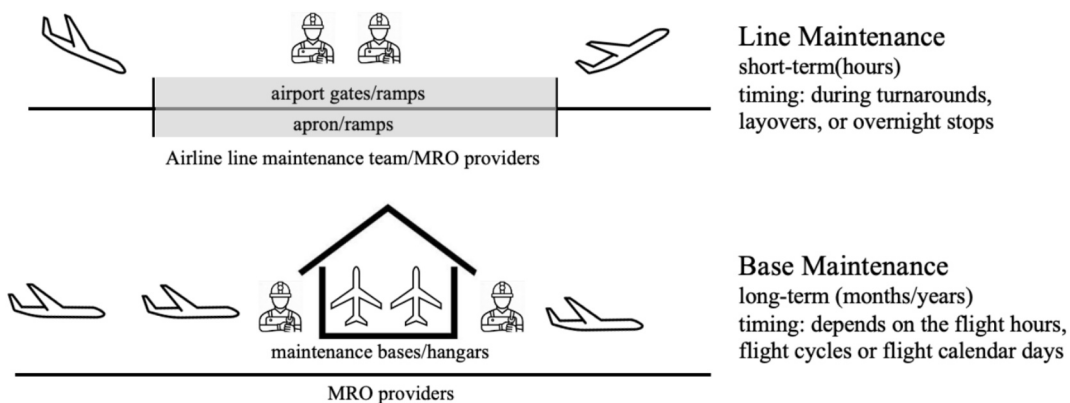


Fig. 3. Diagram of line maintenance and base maintenance.

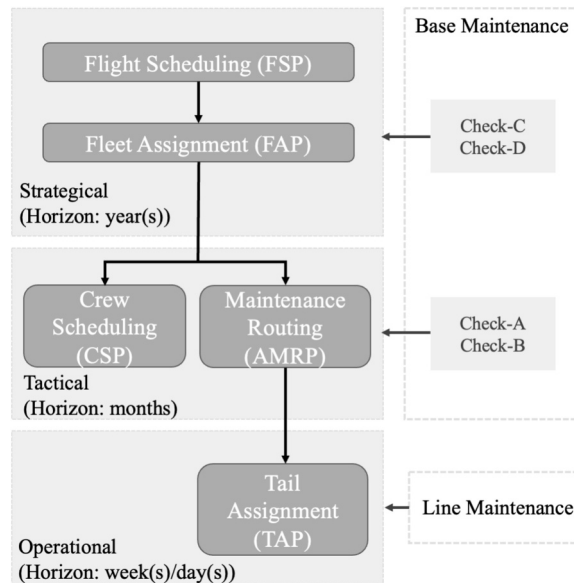


Fig. 4. Overview of maintenance scheduling across different decision-levels.

system component replacements. However, with advancements in modern aviation technology—particularly improved aircraft reliability, onboard diagnostic systems, and enhanced line maintenance capabilities—the B-check as a distinct maintenance category has been largely phased out by most airlines. B-check tasks have either been consolidated into C-checks or delegated to routine A-checks or line maintenance, a shift reflected in contemporary maintenance planning and decision support systems (DSS) (Deng et al., 2021).

In contrast, C-checks and D-checks represent higher-intensity maintenance categories (often referred to as Base Maintenance). C-checks involve systematic disassembly, inspection, and restoration of airframe structures, electrical systems, hydraulic systems, and avionics. D-checks, the most thorough maintenance type, entail near-complete aircraft disassembly and reassembly, involving extensive component replacements, corrosion repairs, and structural assessments. These require meticulous planning and resource integration, with D-check frequency and operational windows typically locked into long-term schedules years in advance to minimize impacts on flight operations (Deng & Santos, 2022; Weide et al., 2022).

Unlike the immediate, fragmented tasks of line maintenance, base maintenance emphasizes systematic, grouped operations following strict logical sequences (e.g., opening, inspection, repair, and sealing) and adheres to highly standardized safety controls and process management (Deng et al., 2021). Task bundling and schedule optimization have emerged as research focal points, with algorithms accounting for operational requirements such as resource availability, task dependencies, duration, and technician skill matching (Chen et al., 2017). These maintenance activities impose significant demands on facilities and resource allocation, typically conducted at airline-owned or outsourced maintenance bases equipped with hangars, heavy lifting equipment, non-destructive testing instruments, specialized tools, and spare parts inventories. Personnel primarily consist of licensed technicians covering multiple disciplines, including structures, electrical systems, avionics, and power systems (Chen et al., 2017). Furthermore, Gopalan (2014) highlighted that the location and distribution of base maintenance facilities not only impact maintenance costs but also exerts influence on airlines' overall maintenance strategies.

Despite its critical role in ensuring flight safety and extending aircraft lifespan, base maintenance presents several operational management challenges. First, extensive grounding periods necessarily diminish aircraft availability, thereby imposing requirements on flight scheduling flexibility. Second, maintenance windows must be coordinated with seasonal schedules, lease terms, and material supply chains, increasing scheduling complexity (Qin et al., 2024). Third, uncertainties in resource availability, such as labor shortages or delays in critical component deliveries, can disrupt planned schedules, necessitating highly responsive scheduling mechanisms and DSS (Weide et al., 2022).

To summarize, base maintenance represents one of the most critical, resource-intensive and systematic components of the aviation maintenance ecosystem. In contrast to line maintenance, which concentrates mainly on routine and immediate inspections to guarantee aircraft preparedness for short-term flights, base maintenance entails a much broader and more thorough range of tasks. This includes meticulous examinations, substantial repairs, and the complete overhaul of essential systems, typically necessitating prolonged downtime, advanced specialized tools, and a team of highly trained professionals.

3.4. Maintenance scheduling at different decision-levels

Aircraft maintenance planning is structure across a hierarchical decision-making framework, ranging from strategic, tactical to operational scheduling, as shown in Fig. 4. These decisions are characterized by distinct planning horizons, fundamentally differentiated by their decision goals, granularity of resource allocation and the degree of uncertainty involved. Specially, strategic decisions

focus on long-term fleet capacity, tactical levels address mid-term maintenance slotting, and operational levels manage real-time scheduling and disruptions.

3.4.1. Strategic level (Long-term)

Strategic-level decisions typically spanning years focus on long-term planning and resource allocation, such as aircraft type selection, the geographical distribution of maintenance bases (Gopalan, 2014), and the scheduling of heavy maintenance cycles such as C-checks and D-checks. Strategic decisions extend beyond fleet maintenance and upgrades to include considerations for expanding maintenance facilities globally, such as establishing additional base maintenance sites to address growing maintenance demands. Es Yurek (2024), suggested that strategic decisions should optimize maintenance resource allocation to reduce maintenance cycles, extend aircraft service life, and thereby lower long-term maintenance costs while ensuring regulatory compliance. Within this context, Adimonyemma & Sun (2025) addressed strategic-level aircraft maintenance planning by proposing an integrated optimization model for large-scale long-term maintenance scheduling and station assignment. Their formulation jointly considers requirements for various maintenance checks, station capacity constraints, and aircraft availability over an extended planning horizon, enabling coordinated decisions across maintenance bases. By explicitly modeling the interaction between long-term maintenance schedules and station assignment, the study demonstrates how strategic planning can improve resource utilization and cost efficiency at the system level.

From a modeling perspective, long-term formulations often incorporate PdM technologies and RUL data to forecast component degradation years in advance. However, the inherent complexity of this scale requires handling high uncertainty (such as market demand fluctuations and equipment aging), which often necessitates multi-objective optimization formulations. A common limitation in the literature is that they often simplify the dynamic adjustment mechanism and limits its applicability in rapidly changing environments.

3.4.2. Tactical level (Medium-term)

Tactical-level decisions spanning on a weekly to monthly planning horizon, address medium-term resource scheduling and task allocation, such as aligning flight schedules with maintenance tasks, including A-checks and B-checks. These decisions aim to optimize resource utilization, ensuring maintenance tasks are completed within constrained time and space windows without disrupting regular flight operations. Unlike strategic decisions, tactical decisions require detailed planning of maintenance activities to manage the trade-off between maintenance tasks and flight scheduling, particularly as fleet sizes grow or flight schedules become more complex. Başdere & Bilge (2014) highlighted tactical decisions must balance aircraft maintenance cycles (e.g., A/B-checks) and resource availability while making trade-offs between resource scheduling and maintenance compliance.

Medium-term models are advantageous as they can integrate multi-day operational data and capture cyclical patterns. For example, Ben Ahmed et al. (2017) optimized fleet allocation and maintenance routing through weekly scheduling, balancing flight continuity and maintenance compliance. The primary challenge at this scale is the “double-sided constraint”: the model must account for short-term operational disturbances while adhering to long-term legal limits (such as legal flight time limits or equipment aging prediction).

3.4.3. Operational level (Short-term)

Operational-level decisions generally spans single to several days of scheduling focus on real-time decision and dynamic adjustment in daily operations, emphasizing immediate resource scheduling, task execution, and contingency responses. These decisions address urgent and routine maintenance tasks, particularly line maintenance activities such as pre-flight/pose-flight inspections, tire replacements, and fluid replenishment. Operational decisions demand high flexibility to address real-time issues like flight delays or equipment failures. The primary goal is to complete maintenance tasks within tight timeframes while ensuring on-time flight departures. Unlike the larger scale, planned tasks at the tactical level, operational decisions prioritize immediate maintenance, such as rapid responses in terminal areas to ensure aircraft readiness for subsequent flights. Vink et al., (2020) emphasize that operational decisions focus on rapidly allocating maintenance resources and addressing tasks in highly uncertain environments to ensure seamless flight operations. Additionally, Jiang et al. (2025) consider operational decisions over a one-day planning horizon, focusing on the coordination of maintenance-related activities to ensure aircraft availability and support uninterrupted flight operations, while allowing maintenance feasibility to be preserved through alternative airport assignments when disruptions render original schedules infeasible.

In terms of optimization, short-term formulation supports dynamic rescheduling and uncertainty management. They must be highly integrated with the flight plan to resolve resource conflicts within time windows. While these models offer superior adaptability and swift response to disruptions. The limitation of short-term formulations is the narrow vision, as they may ignore the impact of short-term decisions on long-term resource availability.

With the emergence of PM and rolling optimization technologies in recent years, the complexity of temporal modeling for maintenance scheduling has markedly increased. PM predicts failures through RUL and sensor data, optimizes short-term and long-term maintenance schedules, and reduces unplanned downtime. For example, the deep learning model can predict engine degradation through real-time data analysis and guide the dynamic adjustment of route maintenance. Rolling optimization refines the scheduling plan through successive iterations, adapts to short-term disturbances (such as flight cancellation) and maintains the stability of long-term plan. These methods require the model to dynamically switch between different time scales and integrate random components (such as uncertain task duration) to improve robustness. However, the computational complexity of PM and rolling optimization is substantial, which requires support from efficient algorithmic approaches, such as approximate dynamic programming or column generation.

The choice of decision-level plays a crucial role in AMS optimization, but its modeling remains challenging, especially in the following aspects. First, there is a lack of standards for modeling goals and constraints at different decision-levels. For example, the dynamic constraints of short-term models are often not effectively integrated with the strategic goals of long-term formulations. Secondly, accurate data support, such as real-time sensor data or supply chain dynamics, is still insufficient for the application of PM and rolling optimization. In addition, there is a lack of modeling considering time scale transition (such as short-term to long-term dynamic convergence), which consequently limits the model utility in multi-scale scenarios. Future research should focus on developing a unified modeling framework across different time scales, combining data-driven methods to improve prediction accuracy, and enhancing the comparability and practical value of models through standardized time scale definitions.

In summary, strategic, tactical, and operational decisions each fulfill distinct functions in maintenance task planning, exhibiting notable differences in their strategic goals, imposed operational and regulatory requirements, and planning horizons.

3.5. Decision domains

The scheduling of maintenance tasks and technical personnel represent the dual pillar of maintenance decision-making. Task scheduling involves optimizing the timing, location, and resource allocation of maintenance tasks while satisfying operational and regulatory requirements. This scheduling process comprises diverse tasks from short cycle line maintenance to long cycle base maintenance and must account for task priorities, temporal windows, and inherent operational uncertainties. Technical personnel scheduling focuses on optimizing the shift, location, and task allocation of maintenance personnel under task requirements and skill requirements, emphasizing skill matching, regulatory compliance, and multi-party interest coordination. Therefore, this section further elaborates on these two distinct scheduling domains within the context of AMS.

3.5.1. Task

In the aviation maintenance system, maintenance task scheduling is defined as the optimization problem of sequencing maintenance tasks under temporal, locational and resource dimensions while satisfying various operational and regulatory requirements. This process aims to ensure aircraft operational safety and regulatory compliance while optimizing maintenance costs and resource utilization (Deng et al., 2021; Villafranca et al., 2025). Task scheduling involves not only managing the time windows of tasks but also coordinating flight timetable, maintenance capacity allocation, and technician scheduling (Chen et al., 2017; Seif & Yu, 2018).

From the perspective of task classification, current literature typically categorizes tasks into line and base maintenance tasks, which are differentiated by task scope, resources requirements, and specific time limits. Line maintenance tasks involve short-cycle, high-frequency, lightweight inspections or replacements, such as A-checks and daily pre-flight or post-flight inspections, while base maintenance tasks encompass longer-cycle, lower-frequency, but highly complex C-checks and D-checks (Safaei & Jardine, 2018). Additionally, scholars have classified tasks based on temporal attributes, distinguishing between periodic and non-periodic tasks, preventive and corrective tasks, and mandatory versus flexible tasks. Tasks are generally outlined at a broad level as aircraft-level maintenance demands, without delving into detailed procedures or skill matching, yet they are logically considered a subset of scheduling tasks.

Priority setting plays a critical role in task scheduling, often determined by factors such as task due time, resource requirements, maintenance duration, and other relevant metrics. Effective prioritization strategies can enhance resource allocation efficiency and increase operational robustness. Moreover, some studies presented the concept of task combination modeling, particularly in scenarios involving modular equipment or complex systems (Levi et al., 2014). This approach emphasized whether to perform other impending or adjacent submodule tasks concurrently with a single task to reduce system maintenance frequency and repeated downtime costs, akin to opportunistic maintenance strategies. In terms of temporal scope, task scheduling spans a wide range, from hourly task window arrangements (e.g., pre-flight and post-flight inspections) to daily, weekly, monthly, or annual long-cycle maintenance planning (e.g., A/B/C-check task bundling) (Witteman et al., 2021). Certain tasks also exhibit flexibility for cross-window scheduling or early execution, requiring coordinated resource allocation and risk assessment across different time scales.

Scheduling optimization has evolved from single-objective cost minimization (e.g., maintenance costs, flight delays, or outsourcing expenses) to multi-objective optimization, incorporating dimensions such as task completion punctuality, resource utilization efficiency, system robustness, and fairness (Chen et al., 2017; Seif & Yu, 2018). Multi-objective optimization has become a prevalent research trend, particularly in the context of heterogeneous tasks and constrained resources. Furthermore, scheduling model paradigms have evolved from static deterministic approaches to dynamic, uncertainty-aware, and robust optimization. For instance, Villafranca et al. (2025) addressed uncertainties associated with task processing times, while He et al. (2023) and Zhang et al. (2024) proposed robust model to enhance system adaptability to disruptions such as flight delays or aircraft swaps. This shift reflects the evolution of maintenance planning from static execution tools to DSS capable of addressing operational disturbances.

3.5.2. Technician

In aircraft maintenance operations, the scheduling of maintenance technicians is one of the most critical components in ensuring the smooth execution of maintenance tasks and the efficient use of resources. Existing research defines technician scheduling as the rational allocation of work shifts, locations, and task assignments for maintenance personnel under given task demands, skill, and operational requirements, aiming to meet maintenance requirements while optimizing cost. This problem encompasses not only technician rostering and assignment decisions but also practical operational requirements such as skill levels, shift regulations, and worker availability.

From a temporal perspective, technician scheduling spans daily, weekly, monthly, or quarterly planning cycles, adapting to the

variability between periodic and ad-hoc maintenance tasks (Yang et al., 2020). For instance, in daily line maintenance, personnel rostering must align rapidly with high-frequency tasks, whereas base maintenance emphasizes long term workforce planning and skill configuration (Ertogral & Öztürk, 2019).

Additionally, Scheduling must precisely match the skill requirements of maintenance tasks, which often demand specific qualification levels (e.g., licenses for technicians) and may require collaborative efforts among multiple technicians. Consequently, skill levels and team configurations represent key considerations for ensuring scheduling feasibility. Research indicates that skill redundancy among technicians can enhance scheduling flexibility but may also increase training costs and reduce skill utilization efficiency, necessitating dynamic balancing through structured configurations and rolling training mechanisms (De Bruecker et al., 2018). Technician scheduling must account for operational requirements, including work locations (e.g., terminals or hangars), daily work-hour limits, rest and shift cycle regulations, and compliance requirements. For example, some studies incorporated requirements such as maximum daily work hours, minimum rest intervals between workers and tasks, and feasible time windows for task transitions into rostering models (Eltoukhy et al., 2022). In outsourced MRO scenarios, technicians from different providers exhibit heterogeneity in skill levels, workforce availability, and labor costs, requiring scheduling strategies to incorporate cross-organizational coordination based on flight routes, maintenance site distributions, and human resource capacities (Eltoukhy et al., 2018b).

Regarding scheduling targets, early studies predominantly focused on labor costs reduction and task completion punctuality (Ertogral & Öztürk, 2019; Qin et al., 2020). More recent research has expanded to include goals such as workload fairness, robustness for robustness (De Bruecker et al., 2015), long-term skill structure evolution (De Bruecker et al., 2018), and multi-stakeholder interest coordination, reflecting a trend toward diversified scheduling goals. Consequently, technician scheduling is no longer positioned as an isolated workforce allocation problem but as an integral component of a broader maintenance-operation-resource coordination system, evolving toward greater systematicity, collaboration, and adaptability.

In summary, the technician scheduling problem has evolved from simplistic static matching models to intricate optimization challenges that incorporate skill structures, rostering regulations, operational uncertainties, and outsourcing coordination. Contemporary research places significant emphasis on designing decision support systems tailored for practical maintenance operations, with a focus on enhancing scheduling feasibility, ensuring fairness, and improving robustness.

3.6. Integrated maintenance and airline scheduling

In airline decision-making, recent research reflects a growing trend toward integrating the AMS with flight scheduling, fleet assignment and crew scheduling. This transformation is driven by the increasing demands for efficient, punctual, and reliable aviation planning in modern aviation systems, coupled with stringent regulatory and economic pressures. These factors compel operational plans to balance maintenance feasibility with economic viability. Maintenance decisions are no longer solely about ensuring safety but have become critical factors influencing overall operational efficiency and market competitiveness. For instance, maintenance scheduling and the emergence of unscheduled maintenance tasks directly affect flight schedule feasibility, while flight delays or crew reassignments affect maintenance task arrangements. Consequently, optimizing AMS decisions in isolation is no longer sufficient to meet operational demands in the context. Integrated maintenance, which involves unified modeling and optimization of maintenance with other airline planning decisions, has emerged as a significant trend in aviation scheduling (Eltoukhy et al., 2019). In integrated contexts, the problem is often framed as the Aircraft Maintenance Routing Problem (AMRP), which involves inserting maintenance events into aircraft flight routes to meet periodic maintenance requirements while minimizing maintenance costs and downtime losses (Ben Ahmed et al., 2017).

Before analyzing integrated maintenance problems, it is essential to clarify the definitions and goals of related airline planning problems which are demonstrated in Fig. 3. Flight Scheduling problem (FSP) determines the timing and frequency of flight segments based on market demand, aiming to maximize revenue or resource utilization (Faust et al., 2017; Cadarso & Vaze, 2023). Fleet assignment problem (FAP) focuses on assigning specific aircraft types to flights, typically to optimize aircraft utilization and reduce operational costs (Özener et al., 2017). Crew scheduling problem (CSP) involves assigning legal and cost-optimal crew rotations to flights, aiming to minimize labor costs while complying with labor regulations. Tail assignment problem (TAP) further assign individual aircraft (by tail number) to performs specific flights, enhancing scheduling precision and operational continuity (Liang et al., 2015). Additionally, maintenance resource allocation addresses the effective scheduling of internal resources, such as maintenance base locations, technician rostering, and spare parts management, to enhance maintenance efficiency (Gopalan, 2014; Ertogral & Öztürk, 2019).

The AMRP, as a standalone problem, has been extensively studied, especially focusing on aircraft flight-hour limitations and detailed task-level management (Başdere and Bilge, 2014; Eltoukhy et al., 2018a,b; Saltzman & Stern, 2022; Qin et al., 2024). However, the problem complexity increases significantly when integrating AMRP with other problems. Integration with a single subproblem is common, with the most typical being integration of AMRP and FSP, which aims to select appropriate aircraft types for flight routes while arranging maintenance schedules to comply with required maintenance regulations. It is particularly relevant for airlines operating multiple aircraft types across complex networks with multiple maintenance bases, requiring optimization of aircraft utilization while ensuring periodic maintenance. Similarly, integrating AMRP with TAP builds on predefined aircraft type assignments, allocating specific tail-numbered aircraft to flight routes and embedding necessary maintenance windows. This approach ensures maintenance are scheduled within each aircraft's flight schedule, enhancing operational continuity and maintenance reliability, particularly in multi-airport scenarios where maintenance base selection aligns with aircraft tail numbers (Liang et al., 2015; He et al., 2023).

The integration of AMRP with CSP aims to coordinate aircraft routes and crew assignments by constructing maintenance-compliant

routes while ensuring legal crew rotations, adhering to labor regulations and flight connectivity requirements. (Díaz-Ramírez et al., 2014). The objective is to synchronize flight and maintenance operations with crew resource management, avoiding disruptions to crew rotations due to maintenance schedules and preventing crew constraints from limiting maintenance feasibility. Shao et al. (2017) highlighted that aircraft routes and crew tasks share highly interdependent flight segment variables, with connectivity consistency constraints posing a significant modeling challenge. To address this, they proposed a decomposition framework based on column generation and pricing mechanisms, achieving coordination among flight segments, aircraft, and crew in scheduling plans. Such models reflect the practical impact of maintenance operations on crew resource management, representing a significant step toward incorporating human resources into AMRP optimization.

When AMRP is integrated with multiple problems, one representative model is the joint optimization of AMRP, FAP, and CSP. This aims to simultaneously allocate aircraft types, flight routes, and crew rotations to achieve optimal resource coordination and maintenance compliance. Ben Ahmed et al. (2022) proposed a robust joint modeling framework incorporating uncertainty to enhance robustness, while Ben Ahmed et al., (2018) emphasized that routes-crew consistency constraints in order to reduce the feasible solution space. Another notable model is the joint integration of AMRP, TAP, and CSP into a unified optimization framework, which involves embedding maintenance nodes into specific tail-numbered aircraft routes while planning crew rotations that comply with labor regulations and flight connectivity requirements. Ruther et al. (2017) employed a branch-and-price algorithm to address pricing subproblems that account for aircraft flight continuity, maintenance compliance, and crew rotation consistency, effectively solving complex constraints. This problem arises in airline operational environments requiring precise tail-number control and high-density flight schedules, addressing multi-objective, multi-constraint challenges involving routes consistency, tail-number binding, and crew work-hour coordination.

Integration of AMRP with multiple problems is common when airlines aim to develop integrated scheduling frameworks, such as those encompassing FSP, FAP, TAP, CSP. These integrated formulations involve a vast number of variables and constraint, rendering the model exceptionally intractable for large-scale instances. Despite challenges, such comprehensive models are essential for accurately reflecting global operational constraints, enhancing global resource utilization efficiency, and reducing operational risks.

3.7. Research trends for maintenance decisions

Based on recent literature, this section identifies key emerging trends and research opportunities across multiple dimensions. In the maintenance strategy, the focus is on innovative methods to enhance service reliability and optimize downtime and resource allocation. Maintenance stakeholders from airlines to MRO providers and collaborative networks is evolving as a whole to address the increasingly complex modern decision-making requirements. At the maintenance categories, priorities and resource allocation are becoming more dynamic, and the perspective of decision-making has shifted towards integrated short-term and long-term planning.

Research into predictive maintenance strategies has increased significantly, marking a transition from traditional CM and PM toward more proactive approaches. This evolution leverages AI and ML to facilitate real-time monitoring of aircraft component health. Prior studies have integrated PdM with deep learning for RUL prediction, reducing unnecessary interventions and improving maintenance resource utilization efficiency (Wang et al., 2024). Current trends utilize AI-powered PdM to forecast failures, minimizing disruptions and extending component life (Cheng et al., 2023). These trends imply the need for more accurate prediction models, and future research could explore hybrid strategies combining PdM with opportunistic maintenance. The development should include enhanced data infrastructures, cost-effective sensors, and sophisticated monitoring tools, as suggested by Aremu et al. (2020). Nevertheless, the field of data management in PdM is under-explored, especially regarding the complexity of heterogeneous data from various aircraft systems. Furthermore, studies on CM, PM, and PdM are widely spread. Limited attention has been paid to opportunistic strategies, leaving a significant gap in the mathematical formulation and algorithmic solutions required for this approach. Moreover, interdisciplinary efforts could be made to integrate maintenance strategies with environmental concerns to achieve emission savings and reduce long-term degradation for both electric and hybrid aircraft fleets.

Airlines and MRO providers have distinct objectives regarding maintenance: airlines aim to optimize aircraft availability and costs for their own fleets, while MRO providers focus on increasing resource utilization efficiency among various airlines. Past work has modeled actor interactions and highlighted conflicts in resource allocation (Qin & Ng, 2023). However, research on collaboration between maintenance entities is rare. Future research could fill this gap by exploring emerging trends, such as blockchain-enabled contracts for transparent data sharing, which enhance trust and coordination amid limited maintenance resources. Additionally, research should prioritize equitable models for small airlines and MROs, as smaller airlines typically possess limited bargaining leverage and receive lower service priority when negotiating with MRO providers.

Line and base maintenance tasks converge with digital tools, blurring traditional boundaries. Studies have shifted A-check tasks to line maintenance for flexibility, using simulation for resource optimization (Deng et al., 2021). Trends feature augmented reality (AR) and drones for real-time line inspections, which are helpful for both line and base operations to reduce downtime. Prior research provides promising findings in cost savings by implementing task grouping strategies (Levi et al., 2014; Golbasi & Turan, 2020; Zhang et al., 2024). Future agendas should investigate integrated task bundling across levels, enabling the flexible scheduling of certain base maintenance tasks into line maintenance at appropriate times, while incorporating robotics for base activities.

Decision-making spans strategic, tactical, and operational planning horizons, with increasing emphasis on adaptability. Earlier models used DP for tactical scheduling under uncertainty (Deng & Santos, 2022). Relatively few studies have developed decision-making models to merge varied planning horizons with diverse airline planning problems. Future research should prioritize hierarchical frameworks capable of synchronizing these temporal horizons, while explicitly accounting for operational uncertainties and disruptions such as adverse weather, unexpected mechanical failures, and maintenance delays.

Task and technician scheduling focus on the coordination of multiple resources and the optimization of task sequences, integrated with priority-based allocation and skills-matching under operational constraints. Regattieri et al. (2015) identify the integration of multiple maintenance operations as a critical research frontier, particularly regarding societal impacts such as workforce upskilling in developing regions. Future directions include data-driven frameworks for matching technician skill levels with specific task requirements, while simultaneously accounting for operational time windows and workforce equity metrics. Workforce training focused on enhancing and refining technicians' skills and expertise is rarely addressed in the literature. However, airlines could significantly enhance operational efficiency by allowing technicians to undergo such training during non-peak or idle periods. Practical innovations are expected regarding ergonomics for healthier work environments, supporting societal goals in aviation.

Additionally, few papers have delved into the management of spare parts in maintenance (Regattieri et al., 2015; Ertogral & Öztürk, 2019). In different scenarios, the maintenance studies have proposed spare parts supply, procurement contracts, and budget planning as key considerations. Nevertheless, the aircraft literature does not provide a detailed coverage of these aspects comprehensively. This leaves an open space for scholars to develop feasible models and efficient solutions.

4. Model formulations and solution approaches

4.1. Problem components and model parameters

The optimization of AMS relies on the rigorous characterization and coordination of objective functions, operational constraints, and planning horizons, which collectively define the mathematical structure and practical applicability. In recent years, the objective formulation has shifted from single objective (such as cost minimization) to multi-objective optimization to balance robustness, punctuality, and passenger experience (Wang et al., 2024). Constraints bare real-world characteristics such as resource limitations, time windows, regulatory requirements, and operational flexibility. Sophisticated modeling framework are required to ensure that scheduling solutions are both operationally feasible and resilient to disruptions. This section discusses the characteristics of objective functions and constraints.

4.1.1. Objective function

The formulation of objective function is a critical step in AMS, as it mathematically quantifies the inherent trade-offs between conflicting operational goals. As shown in Table 4 (appendix), these formulations generally incorporate three dimensions: operation efficiency, economic costs and service quality. These indicators not only measure the cost-efficiency of maintenance plan, but also directly relate to flight safety, regulatory compliance, operation continuity and passenger satisfaction. Early studies tended to focus on single objective optimization, such as minimizing completion time or maximizing revenue. However, with the increasing aviation management complexity and passenger expectations, research has turned to multi-objective optimization, balancing operational cost, robustness, punctuality and passenger centered objectives. In addition, the introduction of prediction indicators such as RUL facilitates the scheduling method based on state and prediction driven. However, some studies only describe the objectives qualitatively and lack clear quantitative indicators, which restrict comparative benchmarking and reproducibility of the models. This section provides a detailed review of the objective function formulations employed in recent AMS optimization studies.

The operational efficiency focuses on the resource-utilization aspect of maintenance activities to maximize aircraft availability and ensure operational continuity. Early studies often aim to reduce total maintenance time by optimizing task sequencing. Other efficiency indicators include maximizing aircraft utilization and minimizing remaining flight time before reaching the statutory maintenance limit. These indicators are particularly critical because they need to quickly complete inspection and maintenance within a short time window to avoid flight delays. For example, the A-check of airline maintenance usually needs to be completed during turnaround period, involving liquid replenishment or preliminary fault diagnosis, requiring efficient task allocation and resource coordination. In base maintenance, the efficiency index focuses on task bundling and hangar utilization optimization to reduce aircraft parking time (Qin et al., 2018). However, some studies only describe the efficiency objectives in general and lack of specific mathematical expression. For example, the weight or priority of completion time is not clearly defined.

The economic costs dimension aims to minimize the operational costs directly or indirectly related to maintenance activities, which remains the primary motivation for airline schedule optimization. The traditional single objective model usually aims at minimizing operational costs, including maintenance-related flight operations costs, labor cost, spare parts inventory cost and outsourcing fees. For example, the optimization of technician scheduling can reduce overtime costs by reasonably arranging shifts, while the optimization of spare parts management can reduce inventory holding costs (Qin et al., 2018). In addition, cost indicators also include indirect costs caused by maintenance, such as penalty costs for flight delays, cancellations or plan changes (Liang et al., 2015). For example, the maintenance window compression caused by delay may result in additional labor or equipment costs, which need to be alleviated by real-time rescheduling.

Service quality has gained prominence in recent AMS models, reflecting a growing emphasis on passenger experience and market competitiveness. Service quality is typically quantified by the frequency of schedule perturbations, including flight and maintenance task cancellations, excessive aircraft ground time and deviation from the baseline maintenance schedule. For example, modifications to the maintenance schedule can propagate to flight delays or cancelations, causing passenger travel interruptions and a significant reduction in satisfaction (Cadarsó & Vaze, 2023). Service quality is also closely related to passenger-center goals, such as optimizing the continuity of passenger travel or reducing flight cancellations caused by maintenance (Ben Ahmed et al., 2017; Cadarsó & Vaze, 2023). Recent studies have introduced prediction indicators based on RUL to allocate maintenance opportunities reasonably by predicting component failures and reduce unplanned downtime and passenger disruption. For example, RUL related model can predict

engine failure through sensor data and schedule maintenance in advance to avoid flight cancellation. However, the quantification of service quality indicators is still insufficient. Some studies only address passenger satisfaction qualitatively, without providing formal mathematical expressions or weight parameters, preventing standardized comparison and limiting the reproducibility of their results.

Driven by increasingly stringent regulatory requirements and the need to balance diverse operational goals, research has shifted from single objective optimization models toward multi-objective optimization, simultaneously incorporating multiple dimensions, such as operational costs, schedule robustness, punctuality and passenger experience. For example, a MILP model with multiple objectives was developed in (Sanchez et al., 2020). Similarly, Cadarso & Vaze (2023) incorporate the continuity and punctuality of passenger travel into their model formulation by integrating maintenance scheduling with flight planning. In multi-objective optimization, especially when balancing monetary profits and costs against non-monetary metrics like delays and satisfaction, assigning appropriate weights is essential to reflect the relative importance of each dimension. For example, minimizing the operational cost often leads to the decrease of punctuality rate. Consequently, Pareto frontier analysis is frequently employed to identify non-dominated, allowing decision-makers to evaluate trade-offs between these conflicting objectives. Predictive indicators like RUL increase model complexity, requiring the optimization process to manage the trade-offs between maximizing a component's life and ensuring flight safety. However, some studies fail to justify their weight selection or priority assignments between different dimensions, making it difficult to interpret their results or compare them with other models.

Although significant progress has been made in the formulation of objective function, there still exists unsolved challenges. First, many studies only qualitatively describe objectives, such as improving passenger satisfaction or reducing plan changes, lacking quantifiable mathematical expression, which limits the benchmark comparison and reproducibility of the model. Service quality indicators often do not provide a rigorous definition of penalty weights or quantitative standards of passenger experience. Secondly, the single objective model lacks flexibility in dealing with dynamic operating environments, such as flight delays and sudden maintenance), while the computational time of the multi-objective model increases rapidly with problem scale due to its inherent complexity, thereby necessitating more efficient algorithms.

4.1.2. Constraints

Constraints in the aviation maintenance scheduling models reflect the operational and regulatory requirements imposed by MRO providers and the complex regulatory environment of aviation operations. These constraints are crucial to ensure the feasibility, robustness and compliance of the maintenance plan, covering resource availability, temporal requirement, capacity and compatibility, operational flexibility, regulatory requirements and other aspects, as shown in Table 5 (appendix). Each type of constraints represents a specific operational challenge in AMS; altogether, they define the mathematical structure of the optimization model.

Resource constraints are central for the maintenance scheduling model, incorporating technician availability, labor capacity, skill levels and shift arrangements (Safaei & Jardine, 2018; Ertogral & Öztürk, 2019). These constraints require precise coordination of human resources to meet maintenance requirements. For example, the availability of technician is limited by the number of employees, shift regulations, and operation priorities, and task allocation needs to be optimized through dynamic labor balance (Borsato Pinhão et al., 2022). The multi skill rostering is crucial because maintenance tasks usually require specific qualifications, such as avionics or structural maintenance certification, which is significantly different between line maintenance and base maintenance. For example, line maintenance tasks may require only one technician with basic skills, while base maintenance C-checks may involve a team of certified engineers in multiple fields. The dynamic labor balance, which involves reallocating technicians in real time to address sudden failures, further increases the complexity of the scheduling model. In addition, resource constraints also involve training costs and skills redundancy. Excessive specialization may reduce flexibility, while insufficient skills may lead to operational inefficiency (De Bruecker et al., 2018). Some studies do not explicitly model these dynamic characteristics, which limits the scalability of solutions in scenarios involving heterogeneous labor for scenarios in outsourced maintenance providers.

Temporal constraints are the core elements in the model, specifying the task sequence within the tight operational windows of airlines. These constraints explicitly characterize maintenance opportunities, flight timetables, residual legal flight time, periodic checks and specific task deadlines (Sanchez et al., 2020). For example, line maintenance tasks, such as liquid replenishment or fault diagnosis, should be executed in the short period during the turnaround time. In contrast, base maintenance tasks require several days or weeks of dwell time and need to be accurately aligned with the flight plan to reduce interference. Residual flight time limits (such as the maximum flight hours specified by the aviation authority) add hard constraints to the scheduling model, while periodic inspections introduce long-term planning requirements (Sanchez et al., 2020). Flight delay or cancellation may lead to maintenance interruption, which further aggravates the complexity of time constraints. However, some studies only address these time dependencies abstractly, lacking explicit mathematical formulations, which limits their application in real-time scheduling systems.

Capacity and compatibility constraints ensure that the maintenance plan is aligned with the physical and logistics capabilities of the facility, including hangar space, workstation restrictions, airport or workshop compatibility, and spare parts inventory. For example, hangar space is one of the bottlenecks in base maintenance. Hangar capacity limits the number of aircraft that can be repaired simultaneously, requiring the utilization rate to be optimized through task bundling and sequencing. Workstation compatibility constraints are only available at some maintenance bases when special equipment is required for specific tasks. Similarly, airport specific constraints, such as limited ground support equipment at international hubs or limited maintenance periods, affect the feasibility of maintenance plans. Spare parts availability adds another layer of complexity, and supply chain delays may interrupt planned maintenance, especially unscheduled maintenance. These constraints require the model to consider resource interdependence and logistics bottlenecks, but the literature often describes these constraints in an implicit or high-level way, lacking detailed mathematical expression, which limits their integration in the integrated scheduling framework.

Some papers support task preemption in their model which allows suspension of low priority tasks in order to accommodate

emergency maintenance tasks, such as critical faults found in inspections. There are constraints which support recovery operations, such as rescheduling tasks or reallocating resources in case of flight delays, cancellations or sudden maintenance needs. For example, flight delays may shorten the time window of maintenance, which requires rapid rescheduling to avoid cascading delays. These constraints require the model to incorporate dynamic decision-making mechanisms, such as rolling horizon optimization approach or real-time recovery algorithms.

Regulatory constraints are imposed by aviation authorities to ensure compliance with safety and airworthiness standards, including periodic inspection, maximum flight hour limit and specific requirements. For example, regulations may stipulate that aircraft must be inspected before reaching a specific cycle limit, imposing hard constraints on the scheduling model. Labor regulations, such as maximum working hours or mandatory rest time for technicians, further complicate personnel scheduling. Some studies also include safety driven constraints, such as risk assessment of key components, to prioritize the task of reducing operational risk. Although regulatory constraints are crucial, they are seldom fully involved in the model and only implicitly mentioned. This opacity hinders the development of standardized models applied across different regulatory environments.

Although significant progress has been made in modeling resource requirement, temporal, capacity and compatibility, and regulatory constraints, several challenges remain in this area. Much of the existing research lacks explicit mathematical formulations, and it is difficult to incorporate the general or implicit constraint description into the framework. For example, temporal constraints (such as maintenance time window restrictions) are frequently discussed, but their alignments with flight timetables often lack sufficient details to capture the real-world operational dynamics. Similarly, capacity constraints are frequently simplified through deterministic assumptions, such as fixed maintenance capacity, while uncertainties in the supply chain, including variability in spare parts availability or delivery schedules, are often overlooked, thereby compromising the adaptability and reliability of the optimization models. The qualitative treatment of operational flexibility and regulatory constraints further restricts the reproducibility and benchmark comparison of the model. To bridge these gaps, subsequent work could concentrate on developing comprehensive and well-defined mathematical models to capture the complexity of constraints. In addition, integrating data-driven methods can enhance constraint robustness through real-time uncertainty modeling. Standardized constraint definition across research will also promote meaningful comparison and enhance the applicability of AMS model in different operating environments.

4.2. Optimization models

AMS relies on different optimization formulations to coordinate complex maintenance tasks, resource allocation, and operational constraints to ensure flight safety, regulatory compliance, and operational efficiency. Table 3 in the appendix demonstrates the classes of optimization formulations covered in this study: deterministic models, stochastic models, and dynamic models. Deterministic models, with their well-defined structure, are widely used in maintenance scheduling problems with fixed parameters, providing accurate planning solutions. Stochastic models introduce uncertainties such as task duration and demand fluctuations to construct a more robust decision-making framework that adapts to operational disturbances in reality. Dynamic models address complex temporal dependencies and dynamic adjustment requirements through sequential decision-making and state dependent modeling, which are particularly suitable for multi-stage and evolving maintenance scenarios. This section explores the characteristics and methodology of deterministic, stochastic, and dynamic optimization models.

4.2.1. Deterministic model

Deterministic optimization models have been extensively adopted in AMS due to their structural clarity, computational efficiency, and the availability of sophisticated solution methodologies. Deterministic models assume that all crucial parameters, including maintenance demand, task durations, resource availability, and operational constraints, are fully known and stable throughout the planning horizon. Under these assumptions, deterministic models formulate the maintenance scheduling problem into well-defined mathematical structures, providing precise and actionable scheduling outcomes. Common deterministic modeling is formulated as IP, MILP, and Mixed-Integer Nonlinear Programming (MINLP), each suitable for different practical contexts and problem complexities. We summarize related literature in Table 3 (Appendix).

In the literature, deterministic optimization models applied to aircraft maintenance broadly fall into three distinct scenarios. The first scenario involves standalone maintenance scheduling problems, which explicitly focus on allocating and timing maintenance tasks independently of other operational decisions. [Witteman et al. \(2021\)](#) uniquely developed the maintenance task allocation problem variant, effectively mapping maintenance tasks to predetermined maintenance windows. Similarly, [Shaukat et al. \(2020\)](#) established a MIP model and presented both exact and heuristic approaches to systematically minimize task deviations from planned schedules, addressing both earliness and tardiness penalties. [Qin et al. \(2020\)](#) further enhanced standalone scheduling by developing a two-stage deterministic framework that optimized hangar task assignments during the initial phase and technician assignments in subsequent phase, effectively addressing resource distribution and enhancing labor utilization.

The second scenario integrates maintenance scheduling explicitly with core airline operational decisions, including FSP, AMRP, FAP, and CSP. Here, maintenance scheduling decisions directly influence operational efficiency and passenger service levels. For instance, [Cadarso & Vaze \(2023\)](#) integrated maintenance with airline schedule and tail assignment decisions, emphasizing passenger itinerary optimization and strict adherence to regulatory maintenance windows. [Ben Ahmed et al. \(2022\)](#) similarly employed an integrated MILP model addressing FAP, AMRP, and CSP simultaneously, balancing profit maximization against operational robustness. [Munari & Alvarez \(2019\)](#) developed a deterministic model that combined aircraft routing with flexible maintenance event scheduling and service upgrades, significantly reducing overall operational costs. Furthermore, [Shao et al. \(2017\)](#) presented a MILP model combining FAP, AMRP and CSP, optimized via Benders decomposition, demonstrating how deterministic models can handle complex,

integrated airline scheduling problems efficiently.

In contrast, the third scenario positions maintenance predominantly as a set of regulatory or operational constraints embedded within broader aviation decision-making frameworks. Unlike the integrated approaches, maintenance scheduling in these models is not reflected in the decision-making variables here but constraints that operational schedules must follow. For example, [Saltzman & Stern \(2022\)](#) utilized a compact multicommodity flow to formulate AMRP, incorporating cyclic maintenance constraints, optimizing costs indirectly through maintenance feasibility conditions. Similarly, [Maher et al. \(2018\)](#) embedded look-ahead maintenance constraints into daily tail assignment problems, explicitly accounting for subsequent days' maintenance requirements to enhance robustness against operational uncertainties. [Díaz-Ramírez et al. \(2014\)](#) also adopted overnight maintenance station constraints in a combined maintenance routing, scheduling, and crew scheduling model, ensuring strict regulatory compliance and operational feasibility without explicitly optimizing maintenance tasks themselves. Such deterministic models illustrate how maintenance constraints, though secondary, profoundly determine the feasibility and resilience of the final solutions.

Collectively, deterministic optimization models provide versatile frameworks capable of addressing diverse contexts within AMS. However, their inherent assumption of parameter certainty also restricts the practical application, prompting increasing research attention towards stochastic and dynamic optimization models, as discussed in subsequent sections.

4.2.2. Stochastic models

While deterministic models provide valuable insights under idealized, known conditions, they are often inadequate for capturing the inherent uncertainty prevalent in real-world aircraft maintenance operations. Stochastic models extend the deterministic paradigm by explicitly incorporating randomness in some parameters, including task durations, component degradation, demand variability, and operational disruptions. By accounting for such uncertainties, stochastic models enable a more robust and risk-aware decision-making framework.

A significant strand of research leverages two-stage stochastic programming (TSSP) to account for uncertainty in scheduling. [Villafranca et al. \(2025\)](#) formulated a TSSP model that incorporates uncertain maintenance task durations to optimize line maintenance scheduling. The first-stage decisions involve task outsourcing and sequencing, while the second stage adapts task timings and flight departures based on realized scenarios, with expected costs estimated via Sample Average Approximation (SAA). The authors employed both an exact Integer L-shaped decomposition and a tailored Adaptive Iterated Local Search (AILS) heuristic for tractability for large scale instances. Similarly, [Patriksson et al. \(2015\)](#) investigated the opportunistic replacement in multi-component systems, modeling component failure times using Weibull distributions. Their integer L-shaped method enhances solution efficiency by tightening lower bounds in the recourse function. [Liu et al. \(2019\)](#) introduced a risk-averse TSSP model for fleet assignment under demand uncertainty, integrating the Conditional Value-at-Risk (CVaR) to manage downside risk. In the initial phase, aircraft groups are allocated to every flight route, whereas in the subsequent phase, specified aircraft types are assigned guided by more accurate demand forecasts. Their results demonstrate superior robustness in fleet planning, particularly under volatile market conditions and high-risk aversion.

Beyond traditional two-stage structures, multi-agent and hierarchical decision settings have been incorporated through stochastic bilevel optimization. [Eltoukhy et al. \(2018b\)](#) introduced a Stackelberg game-based stochastic MILP where airlines (leader) are responsible for aircraft routes and maintenance, while the MRO providers who act as followers are responsible for technician staffing plans. Their framework captures uncertainty in task occurrence and duration and is solved via ant colony optimization (ACO) algorithm approach, showing improved coordination and resource utilization over decoupled planning schemes.

Complementary research leverages simulation-based stochastic modeling to capture system dynamics in the presence of complex, interdependent sources of uncertainty. [Varennna et al. \(2025\)](#) presented a stochastic discrete event simulation (SDES) framework incorporating randomness in task processing times, unscheduled maintenance events, and resource conflicts. [De Bruecker et al. \(2015\)](#) also employed enhanced optimization in personnel scheduling, proposing a model enhancement heuristic that iteratively integrates empirical stochastic aircraft arrival delay distributions into a MILP-based scheduler. Their approach outperforms naive buffer-based policies by achieving superior reliability under stochastic arrival delays.

Another crucial application of stochastic modeling lies in PM and prognostics, where component-level uncertainty plays a central role. [Niu et al. \(2023\)](#) introduced a nonlinear state-space degradation model capturing both intrinsic and interaction-induced degradation in multi-component systems. Their method integrates Kalman filtering with Expectation-Maximization to estimate hidden degradation states and RUL, with results highlighting the importance of modeling stochastic dependence in improving RUL accuracy.

Finally, the interface between SP and dynamic decision-making is explored in the work of [Deng & Santos \(2022\)](#), who propose a lookahead approximate dynamic programming (ADP) algorithm. By modeling aircraft elapsed time and maintenance duration as stochastic processes, the authors formulate a Markov Decision Process (MDP) and make schedules for long-horizon checks with Monte Carlo simulations for short-horizon adaptability. Their approach reduces unnecessary maintenance slots while preserving high utilization, thereby exemplifying how stochastic sampling can enhance dynamic policy design.

In summary, stochastic models have proven indispensable for incorporating real-world uncertainty into AMS.

4.2.3. Dynamic models

Dynamic optimization models explicitly consider sequential decision-making processes, where maintenance decisions are periodically revised in response to evolving system states and newly acquired information. In contrast to deterministic models, which assume all future parameters to be known with certainty, and stochastic models, which typically employ scenario-based planning without explicitly capturing adaptive decision-making at intermediate stages, dynamic models integrate both uncertainty and adaptive

temporal decisions into a unified sequential decision framework. Central to dynamic optimization is the formulation of the decision-making environment as an MDP, whereby decisions taken at any given stage affect not only immediate costs but also future states, costs, and available decisions. Thus, dynamic models are particularly suited for problems characterized by complex intertemporal dependencies, adaptive resource allocations, and evolving system conditions under uncertainty.

A prominent early application of dynamic models in AMS is presented by [Deng et al. \(2020\)](#) through a structured DP approach. In their work, aircraft maintenance checks are scheduled dynamically to minimize unused flight hours between maintenance events, consequently minimizing required checks and enhancing fleet availability. A recursive DP formulation is employed, systematically traversing state transitions represented by aircraft utilization metrics and maintenance windows. Building on this foundation, [Deng & Santos \(2022\)](#) further developed the aircraft maintenance check scheduling problem by incorporating explicit stochastic characteristics in aircraft utilization and maintenance durations. Their model is articulated as a stochastic MDP, solved via an ADP methodology. The authors use a rolling-horizon approach combined with SAA and embedded MIP formulations for lookahead optimization. This hybrid framework strikes a balance between computational efficiency and modeling accuracy, effectively managing uncertain flight schedules and stochastic task durations.

[Lagos et al. \(2020\)](#) addressed short-term operational maintenance scheduling challenges by integrating tail assignment decisions within a dynamic MDP framework. Recognizing that maintenance tasks are revealed progressively and must be allocated dynamically to constrained aircraft ground-time windows, the authors propose ADP strategies, including rolling horizon methods, value function approximation (VFA), and hybrid VFA-rolling horizon policies. Addressing the complexity inherent to engine maintenance scheduling, [Zhang et al. \(2025\)](#) proposed an advanced dynamic modeling approach that combines stochastic dynamic integer programming (SDIP) with an ADP solution method rooted in Stochastic dual dynamic programming (SDDP). Unlike previous studies, this approach explicitly models endogenous uncertainties, notably the deterioration and failure probabilities influenced directly by selected maintenance policies. By employing a dynamic state-space representation through a graph-based framework and dynamically generating state graphs, the authors effectively manage computational complexity and dimensionality issues.

In conclusion, a range of optimization techniques have been employed for the AMS problem. Dynamic optimization models tend to outperform deterministic and stochastic models by enabling sequential, state-dependent decisions that adapt to evolving system conditions. Their ability to capture intertemporal dependencies and respond to uncertainty makes them particularly valuable in complex, real-world maintenance environments.

4.3. Solution approaches

A diverse range of solution methodologies have been proposed to address the complex multi-objective optimization problems and dynamic nature of AMS. Historically, these approaches have been categorized into exact algorithms and heuristics, though recent advancements have been introduced data-driven methods as a critical third pillar. Exact algorithms are designed to find optimal solutions, making them suitable for scenarios with moderate scale and strict compliance with regulatory constraints. Heuristic and metaheuristic algorithms deal with large-scale and complex problems through genetic algorithms, neighborhood search and other methods, provide efficient approximate solutions, and adapt to dynamic scheduling requirements. Data-driven methods leverage sensor data and historical records, providing the empirical basis for AMS.

4.3.1. Exact algorithms

Exact algorithms are crucial in solving AMS, particularly when achieving optimality is essential and computationally feasible. These algorithms, typically utilize exact techniques such as B&B, B&P and Benders decomposition. For instance, [Shao et al. \(2017\)](#) and [Es Yurek \(2024\)](#) developed Benders decomposition approaches to tackle operational aircraft maintenance routing problems. These approaches typically break down the problem into smaller subproblems (main problem and subproblems), which makes them more tractable. Benders decomposition has proven effective in solving large-scale problems by efficiently handling complex maintenance constraints that are difficult to model in a single optimization problem.

Recent studies in AMS have greatly benefited from advanced decomposition methods like column generation and B&P, which have proven effective in large-scale applications. For example, [Ruther et al. \(2017\)](#) demonstrated the efficacy of B&P for solving integrated problem, where maintenance operations are a critical component. These methods decompose the intricate optimization challenge into more manageable subproblems, simplifying the handling of interdependent factors such as flight routes and maintenance schedules.

[Liang et al. \(2015\)](#) extended this concept by applying two-stage column generation techniques to mitigate the overall expected propagated delay. This study highlights the importance of considering longer planning horizons (weekly instead of daily), as this approach increases opportunities for better scheduling. Column generation works with multiple proposed lower bounds to solve pricing problems, focusing on the most promising variables, which significantly reduces computational effort.

Moreover, maintenance capacity constraints are often incorporated into exact algorithms to account for the limited number of available maintenance resources. [Başdere & Bilge \(2014\)](#) extended their ILP models by integrating remaining flying time in their formulations, in order to minimize unused legal flying time, ensuring that the maintenance schedules are realistic and feasible within the operational limits of the facilities. This addition supports the optimization of maintenance scheduling, taking into account practical constraints, such as the availability of maintenance resources.

While exact algorithms offer optimal solutions, they face significant scalability challenges as problem size increases. [Başdere & Bilge \(2014\)](#) introduced compressed annealing (CA), a heuristic method that provides computationally efficient solutions for large-scale instances where exact methods become intractable. While CA does not guarantee optimality, it has been shown to provide effective solutions within a notably shorter time span, rendering it a practical tool for operational environments that require timely

decision-making.

In summary, exact algorithms remain vital in the field of AMS, especially when the objective is to obtain optimal solutions that adhere strictly to operational constraints. However, limitations in scalability and computational efficiency persist, requiring alternative approaches to address these issues.

4.3.2. Heuristics and metaheuristics

Heuristic approaches have emerged as practical solutions to complex and computationally intensive problems associated with AMS. These methods offer valuable alternatives when exact algorithms become intractable within a realistic time frame due to the scale and intricacy of real-world maintenance scenarios. In the aviation domain, heuristics are critical for ensuring timely maintenance while handling operational disruptions, such as flight delays or resource conflicts. Techniques like genetic algorithms and neighborhood search optimize task sequencing, technician allocation, and resource utilization, balancing efficiency and robustness. For instance, heuristics enable rapid rescheduling in response to unexpected faults, maintaining operational continuity. This section offers a detailed summary of the heuristic techniques proposed in recent literature.

Among the most prominent heuristic strategies are the genetic algorithm (GA) and its hybrids. Patriarca et al. (2025) embedded a GA within a functional resonance analysis framework to optimize maintenance resource allocation, while Chen et al. (2024) used a genetic-based heuristic in a bi-objective project scheduling context to jointly minimize maintenance duration and state changes. These approaches capitalize on the adaptive search capabilities of GA in complex decision spaces. Likewise, heuristic methods based on column generation have been frequently employed for large-scale routing and recovery problems. Liang et al. (2018) applied column generation to handle the recovery scenario, accommodating flexibility, while Liang et al. (2015) extended this framework to robust weekly maintenance routing and tail assignment, both leveraging iterative master-subproblem decomposition to handle combinatorial complexity. Faust et al. (2017) and Gürkan et al. (2016) also applied column generation and discretized heuristics, respectively, to integrate demand-oriented scheduling and cruise speed control into operational planning.

A second research cluster focuses on neighborhood search-based approaches, which blend exact optimization with heuristic refinement. Ben Ahmed et al. (2022) combined MIP relaxation with proximity search in a metaheuristic method to integrate FAP, AMRP, and CSP. Al-Thani et al. (2016) adopted a MIP-enhanced large neighborhood search for maintenance routing, emphasizing solution-space reduction and iterative improvement. Similarly, Chen et al. (2017) employed a tabu search heuristic for technician-task assignment, balancing labor cost and workload fairness. Game-theoretic and sequential decision heuristics also feature prominently. Yang et al. (2018) proposed a sequential game heuristic with state backtracking for selective maintenance scheduling under phased missions, while Kozanidis et al. (2014) designed iterative heuristics for mission-oriented flight and maintenance arrangement. Within the field of workforce scheduling, Yang et al. (2020) combined simulation with a backward removal heuristic to improve workforce planning in air-cargo operations. Lastly, Gopalan (2014) explored heuristic algorithms such as greedy methods, walk-and-mark procedures, and graph-based formulations like vertex cover and independent set for strategic maintenance base location, analyzing their performance within facility location problems on Eulerian networks.

4.3.3. Data-driven approaches

Data-driven techniques have become pivotal in aviation PdM, providing real-time forecasts for fault detection and maintenance scheduling. By using sensor data, historical maintenance records and operation indicators, these methods help improve the accuracy of maintenance timing and system reliability, minimizing aircraft downtime. Machine learning-related approaches, such as regression and classification models, are excellent at predicting the RUL of components and identifying potential faults from complex data sets. The deep learning methods, including CNNs and LSTM models, capture complex spatio-temporal patterns in sensor data and improves the accuracy of fault prediction. The hybrid statistical model combines traditional time series analysis approaches, such as ARMA, and ML techniques and provides robust prediction by integrating probability and data-driven insights. This section presents various data-driven methods, including ML, DL, and hybrid statistical models, which have been successfully integrated into aircraft maintenance decision-making.

Machine learning approaches

ML techniques have attracted considerable interest and attention in PM because of their capacity to process large datasets and uncover complex patterns. ML-based methods are particularly useful for forecasting the RUL of components and predicting fault events based on historical data. Jordan et al. (2022) studied aircraft performance metrics and ML for predicting availability of the KC-135R aircraft. Their approach involves analyzing aircraft usage data to forecast maintenance needs, ultimately improving scheduling decisions and ensuring operational readiness. This highlights how machine learning techniques, combined with operational metrics, can optimize maintenance planning and resource allocation in military aviation. Beyond centralized learning paradigms, recent studies have begun to explore federated learning (FL) as an emerging branch of ML for predictive maintenance, particularly under data privacy and decentralization constraints. Xu et al. (2025) proposed a federated learning framework that integrates adaptive sampling and ensemble techniques for aircraft engine RUL prediction, enabling collaborative model training across distributed data sources without sharing raw data. Their results show that FL-based approaches can achieve robust and accurate RUL predictions while addressing data heterogeneity and privacy concerns. In parallel, Jiang et al. (2023) introduced the synergy between ML and mixed reality (MR) within the framework of smart aviation systems. By integrating real-time data from aircraft systems with MR interfaces, they propose a system that aids maintenance personnel by providing visualizations of predictive maintenance outcomes. This combination of ML and MR enhances decision-making, improving fault detection and the accuracy of maintenance predictions.

Deep learning approaches

Deep learning methods have shown great potential in modeling complex, high-dimensional data, such as that generated by aircraft

sensors. These techniques, including CNNs and LSTMs, excel at identifying intricate temporal patterns and spatial dependencies, making them ideal for predictive maintenance in aviation. [Cheng et al. \(2023\)](#) developed a DL-based ensemble method to forecast the RUL of aircraft components, combining multiple deep learning architectures. The model efficiently incorporates temporal and spatial relations in the dataset, improving fault prediction accuracy. The ensemble approach helps leverage the strengths of different deep learning models, ensuring robust predictions even in the face of noisy or incomplete data. [Chen et al. \(2025\)](#) proposed a cluster edge-based batch active learning framework integrated with a CNN–BiLSTM–Attention architecture for aircraft engine RUL prediction. This approach improves prediction accuracy while reducing labeling costs, thereby enhancing the practicality of deep learning–based predictive maintenance under data scarcity. These deep learning models are enhanced by advanced feature engineering techniques to ensure the data used for training is both relevant and well-structured. [Aremu et al. \(2020\)](#) formulated a feature selection methodology founded on correlation and relative entropy (C-RE), which identifies and retains the most informative features while reducing redundancies in the data. This feature engineering process is crucial in improving the performance of deep learning models by ensuring they focus on the most relevant input variables.

Hybrid approaches and statistical models

While ML and DL techniques are powerful, integrating these with traditional statistical methods can offer even greater predictive capabilities. [Baptista et al. \(2018\)](#) employed ARMA forecasting in combination with data-driven models for fault event prediction. ARMA refers to a time-series forecasting model, which was used to analyze the temporal relationship in historical fault event data. ARMA predictions are then integrated into machine learning models to provide a more accurate and comprehensive prediction of maintenance needs. [Savitha et al. \(2020\)](#) introduced an unsupervised learning approach that adapts incrementally to new data without retraining from scratch. This model has shown potential in dynamic maintenance scenarios, where real-time predictions are required. The ability to learn online makes the RBM especially useful in maintenance environments that deal with continuously evolving data. [Kraus & Feuerriegel \(2019\)](#) proposed an interpretable deep learning model for RUL forecasting, incorporating Variational Bayesian Inference (VBI) to provide probabilistic interpretations of predictions. This method boosts the transparency of the model, which is critical in safety–critical domains like aviation, where understanding prediction uncertainty is essential. By integrating VBI, the model offers more reliable and interpretable maintenance forecasts and helps maintenance crews to make better-informed decisions. Additionally, [Wang et al. \(2024\)](#) combined DL with MILP models to optimize maintenance scheduling decisions. This hybrid approach incorporates predictive models for fault detection and integrates them with optimization algorithms to minimize downtime and maintenance costs.

Data-driven approaches, including ML, DL, and hybrid models, have significantly advanced predictive maintenance strategies in aircraft. The fusion of classical statistical methods with cutting-edge ML-based strategies has exhibited remarkable effectiveness for fault detection, RUL prediction, and maintenance scheduling. Moreover, methods such as feature selection, ARMA forecasting, and dimensionality reduction further enhance the prediction accuracy and efficiency of these models.

4.4. Research trends for modelling and solution approaches

This section explores emerging research trends in modeling frameworks and solution methodologies. As airlines seek to optimize efficiency and revenue, the development of sophisticated model formulation becomes essential for managing both deterministic and stochastic environments. In the design and management of maintenance decisions, it is critical to incorporate uncertainties in aspects such as flight delays, arrival/departure times, and unexpected tasks to boost overall performance when facing disruptions. While deterministic models are widely studied, more attention should be paid to stochastic or dynamic paradigms to incorporate real-world characteristics, thereby increasing the robustness of model performance and achieving a safer flight experience. Recent evidence also suggests growing acceptance of AI-supported maintenance practices, as passengers tend to view AI-based predictive maintenance positively for its potential to enhance safety and operational reliability ([Fondevila-Gascón et al., 2025](#)).

Cost minimization is prevalent in defining optimization goals, as it provides a practical assessment of maintenance operations in real-life applications. Overly simplistic cost functions might contribute to an inadequate measurement of decisions and strategies. Performance measures could expand beyond cost minimization to include robustness, passenger satisfaction, and sustainability metrics like carbon footprints ([Wang et al., 2024](#)). Integrating sustainability perspectives involving environmental concerns may be promising. Constraints incorporate skill heterogeneity and regulatory compliance, with models balancing multiple objectives ([Eltoukhy et al., 2022](#)). Scholars could exert more effort to make progress toward a low-carbon world with advanced fuels such as SAF to achieve social benefits.

Single-objective formulations have been prevalent for optimization problems and are typically solved using commercial solvers like CPLEX and Gurobi. Nevertheless, a significant gap remains in the development of comprehensive framework capable of synchronizing various planning horizons and different scheduling problems. The development of more efficient and specialized algorithms is required to address these complex models within reasonable computational times ([Es Yurek, 2024](#)). We advocate for the development of exact algorithmic methods specifically designed for multi-objective optimization. Additionally, solution frameworks often incorporate heuristic or metaheuristic methodologies to address computational complexity in maintenance scheduling, offering efficient alternatives to exact algorithms ([Chen et al., 2017](#)). Moreover, blending exact methods (e.g., branch and price, column generation) with heuristics and metaheuristics holds considerable promise. Furthermore, there is a notable trend toward AI-driven techniques (e.g., ML and DL) for real-time problem-solving, including the use of variational inference to achieve interpretable predictions ([Kraus & Feuerriegel, 2019](#)). Future directions should encompass scalable metaheuristics integrated with explainable AI, thereby addressing existing gaps in handling large-scale instances. In addition, practical tools such as cloud-based platforms could democratize access to these methodologies, while offering interdisciplinary opportunities in AI ethics to ensure transparent decision-making and ultimately

foster greater societal trust in automated systems.

5. Conclusion

Decision-making in AMS is critical for aviation safety, operational efficiency, and economic sustainability, and has become a vital and challenging topic in the operations research and management science community. An increasing research trend is promising from the distribution of papers by year, while a comprehensive review regarding AMS is rare in the research community. To this end, this review provides a comprehensive overview of 98 articles published between January 2014 and December 2025 from selected OR/MS journals. We categorize the literature into maintenance decision-making, optimization models, and solution approaches. These insights provide the basis for developing more robust maintenance strategies that can handle the increasing complexity of modern aviation systems.

We identify several trends in AMS that define the current research landscape. There is a clear shift from traditional scheduled maintenance toward PdM to optimize maintenance timing and resource utilization. These trends imply the need for more accurate prediction models and research on opportunistic maintenance. Another critical operational gap lies in the conflicting objectives of different maintenance stakeholders. Airlines prioritize maximum aircraft availability and punctual operations, whereas MRO providers focus on maximizing resource utilization efficiency and profitability. Current research lacks robust collaborative mechanisms that can mitigate these conflicts to foster “win-win” outcomes. We also highlight the potential for integrating task bundling across levels to move certain base maintenance operations into line maintenance, which is promising for decreasing aircraft downtime and improving the utilization rate. Finally, there is a growing trend of integrating the AMS with other planning problems (e.g., FSP, FAP, CSP, AMRP, TAP). This holistic approach is essential to tackle the suboptimality inherent in isolated decisions and for archiving the operational resilience required in modern, complex aviation networks.

For decision-makers to design and to manage maintenance, it is important to incorporate uncertainties that suit real-world situations, provide robust performance and offer a safe flight experience. Furthermore, decision-makers must consider different dimensions, with a shift from single-objective formulations to multi-objective models likely to become prevalent in the future, and we expect research to incorporate sustainability metrics, such as carbon footprints and SAF usage, in these decisions. The continuous growth in passenger and cargo demand has led to increasingly large-scale problem instances, requiring more computationally efficient solution approaches. Exact algorithms could be developed to solve multi-objective problems, along with tailored heuristics/meta-heuristics for large-scale problems. As a prominent research area, the future of AMS lies in AI-integrated solutions that use explainable models to handle large-scale, uncertain environments, along with the operational improvements they provide.

To summarize, this paper reviews the latest literature employing operations research methods to enhance AMS decisions, while updating the cutting-edge insights. Future research directions are outlined. We hope this work establishes a solid groundwork for the field and sparks further investigations in AMS.

CRedit authorship contribution statement

Zimu Guo: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Xin Wen:** Visualization, Supervision, Project administration, Investigation, Conceptualization. **Xinyu Huang:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Authors sincerely thank editors and reviewers for their valuable comments which have helped improved the paper significantly.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tre.2026.104767>.

Data availability

No data was used for the research described in the article.

References

Adimanyemma, J., Sun, Y., 2025. Optimization model for large-scale long-term aircraft maintenance scheduling and station assignment. *Transportation Research Part E: Logistics and Transportation Review* 202, 104302.

- Al-Thani, N.A., Ben Ahmed, M., Haouari, M., 2016. A model and optimization-based heuristic for the operational aircraft maintenance routing problem. *Transp. Res. Part C Emerging Technol.* 72, 29–44.
- Aremu, O.O., Cody, R.A., Hyland-Wood, D., McAree, P.R., 2020. A relative entropy based feature selection framework for asset data in predictive maintenance. *Comput. Ind. Eng.* 145, 106536.
- Baptista, M., Sankararaman, S., Medeiros, D.e., Ivo, P., Nascimento, C., Prendinger, H., Henriques, E.M.P., 2018. Forecasting fault events for predictive maintenance using data-driven techniques and ARMA modeling. *Comput. Ind. Eng.* 115, 41–53.
- Bagdere, M., Bilge, Ü., 2014. Operational aircraft maintenance routing problem with remaining time consideration. *Eur. J. Oper. Res.* 235 (1), 315–328.
- Ben Ahmed, M., Ghroubi, W., Haouari, M., Sherali, H.D., 2017. A hybrid optimization-simulation approach for robust weekly aircraft routing and retiming. *Transp. Res. Part C Emerging Technol.* 84, 1–20.
- Ben Ahmed, M., Zeghal Mansour, F., Haouari, M., 2018. Robust integrated maintenance aircraft routing and crew pairing. *J. Air Transp. Manag.* 73, 15–31.
- Ben Ahmed, M., Hryhoryeva, M., Hvattum, L.M., Haouari, M., 2022. A matheuristic for the robust integrated airline fleet assignment, aircraft routing, and crew pairing problem. *Comput. Oper. Res.* 137, 105551.
- Borsato Pinhão, J.D.M.O., Ignacio, A.A.V., Coelho, O., 2022. An integer programming mathematical model with line balancing and scheduling for standard work optimization: a realistic application to aircraft engines assembly lines. *Comput. Ind. Eng.* 173, 108652.
- Cacchiani, V., Salazar-González, J.-J., 2017. Optimal Solutions to a real-world integrated airline scheduling problem. *Transp. Sci.* 51 (1), 250–268.
- Cadaro, L., Vaze, V., 2023. Passenger-Centric Integrated Airline Schedule and Aircraft Recovery. *Transp. Sci.* 57 (3), 813–837.
- Chen, G., He, W., Leung, L.C., Lan, T., Han, Y., 2017. Assigning licenced technicians to maintenance tasks at aircraft maintenance base: a bi-objective approach and a chinese airline application. *Int. J. Prod. Res.* 55 (19), 5550–5563.
- Chen, G., He, W., Tian, Y., Ma, K., 2024. Resource-constrained project scheduling with multiple states: Bi-objective optimization model and case study of aircraft maintenance. *Comput. Ind. Eng.* 191, 110169.
- Chen, Y., Wang, L., Zhao, X., 2025. Cluster edge-based active learning with batch queries for aircraft engine RUL prediction under label scarcity. *Ann. Oper. Res.* 1–34.
- Cheng, Y., Zeng, J., Wang, Z., Song, D., 2023. A health state-related ensemble deep learning method for aircraft engine remaining useful life prediction. *Appl. Soft Comput.* 135, 110041.
- Dangut, M.D., Skaf, Z., Jennions, I.K., 2022. Handling imbalanced data for aircraft predictive maintenance using the BACHE algorithm. *Appl. Soft Comput.* 123, 108924.
- De Bruecker, P., Van Den Bergh, J., Beliën, J., Demeulemeester, E., 2015. A model enhancement heuristic for building robust aircraft maintenance personnel rosters with stochastic constraints. *Eur. J. Oper. Res.* 246 (2), 661–673.
- De Bruecker, P., Beliën, J., Van Den Bergh, J., Demeulemeester, E., 2018. A three-stage mixed integer programming approach for optimizing the skill mix and training schedules for aircraft maintenance. *Eur. J. Oper. Res.* 267 (2), 439–452.
- Deng, Q., Santos, B.F., 2022. Lookahead approximate dynamic programming for stochastic aircraft maintenance check scheduling optimization. *Eur. J. Oper. Res.* 299 (3), 814–833.
- Deng, Q., Santos, B.F., Curran, R., 2020. A practical dynamic programming based methodology for aircraft maintenance check scheduling optimization. *Eur. J. Oper. Res.* 281 (2), 256–273.
- Deng, Q., Santos, B.F., Verhagen, W.J.C., 2021. A novel decision support system for optimizing aircraft maintenance check schedule and task allocation. *Decis. Support Syst.* 146, 113545.
- Díaz-Ramírez, J., Huertas, J.I., Trigos, F., 2014. Aircraft maintenance, routing, and crew scheduling planning for airlines with a single fleet and a single maintenance and crew base. *Comput. Ind. Eng.* 75, 68–78. <https://doi.org/10.1016/j.cie.2014.05.027>.
- Dinis, D., Teixeira, A.P., Barbosa-Póvoa, A., 2020. ForeSim-BI: a predictive analytics decision support tool for capacity planning. *Decis. Support Syst.* 131, 113266.
- Eltoukhy, A.E.E., Chan, F.T.S., Chung, S.H., Niu, B., 2018a. A model with a solution algorithm for the operational aircraft maintenance routing problem. *Comput. Ind. Eng.* 120, 346–359.
- Eltoukhy, A.E.E., Wang, Z.X., Chan, F.T.S., Chung, S.H., 2018b. Joint optimization using a leader–follower Stackelberg game for coordinated configuration of stochastic operational aircraft maintenance routing and maintenance staffing. *Comput. Ind. Eng.* 125, 46–68.
- Eltoukhy, A.E.E., Wang, Z.X., Chan, F.T.S., Fu, X., 2019. Data analytics in managing aircraft routing and maintenance staffing with price competition by a Stackelberg–Nash game model. *Transportation Research Part e: Logistics and Transportation Review* 122, 143–168.
- Eltoukhy, A.E.E., Wang, Z.X., Shaban, I.A., Chan, F.T.S., 2022. Coordinating aircraft maintenance routing and integrated maintenance staffing and rostering: a Stackelberg game theoretical model. *Int. J. Prod. Res.* 60 (24), 7450–7474.
- Ertogral, K., Öztürk, F.S., 2019. An integrated production scheduling and workforce capacity planning model for the maintenance and repair operations in airline industry. *Comput. Ind. Eng.* 127, 832–840.
- Es Yurek, E., 2024. Combinatorial benders decomposition for the operational aircraft maintenance routing problem. *Comput. Oper. Res.* 164, 106545.
- Faust, O., Gönsch, J., Klein, R., 2017. Demand-Oriented integrated scheduling for point-to-point airlines. *Transp. Sci.* 51 (1), 196–213.
- Fondevil-Gascón, J.F., Gutiérrez-Aragón, Ó., Lopez-Lopez, D., Curiel-Barrios, G., Alabart-Algueró, J., 2025. Passenger perceptions of Artificial Intelligence in airline operations: Implications for air transport management. *J. Air Transp. Manag.* 129, 102874.
- Golbasi, O., Turan, M.O., 2020. A discrete-event simulation algorithm for the optimization of multi-scenario maintenance policies. *Comput. Ind. Eng.* 145, 106514.
- Gopalan, R., 2014. The aircraft maintenance base location problem. *Eur. J. Oper. Res.* 236 (2), 634–642. <https://doi.org/10.1016/j.ejor.2014.01.007>.
- Gürkan, H., Gürel, S., Aktürk, M.S., 2016. An integrated approach for airline scheduling, aircraft fleet and routing with cruise speed control. *Transp. Res. Part C Emerging Technol.* 68, 38–57.
- Gustavsson, E., Patriksson, M., Strömberg, A.-B., Wojciechowski, A., Önnheim, M., 2014. Preventive maintenance scheduling of multi-component systems with interval costs. *Comput. Ind. Eng.* 76, 390–400.
- He, Y., Ma, H.-L., Park, W.-Y., Liu, S.Q., Chung, S.-H., 2023. Maximizing robustness of aircraft routing with heterogeneous maintenance tasks. *Transportation Research Part e: Logistics and Transportation Review* 177, 103237.
- Hu, Y., Miao, X., Zhang, J., Liu, J., Pan, E., 2021. Reinforcement learning-driven maintenance strategy: a novel solution for long-term aircraft maintenance decision optimization. *Comput. Ind. Eng.* 153, 107056.
- Jiang, Y., Tran, T.H., Williams, L., 2023. Machine learning and mixed reality for smart aviation: applications and challenges. *J. Air Transp. Manag.* 111, 102437.
- Jiang, J., Zhang, S., Tang, Y., Guo, Y., Wu, C.L., 2025. ADMM-based augmented Lagrangian methods for robust aircraft recovery problem considering connection time, resource capacity and maintenance flexibility. *Transportation Research Part e: Logistics and Transportation Review* 201, 104243.
- Jordan, J.A., Etemadi, A., Grenn, M.W., 2022. Predicting KC-135R aircraft availability with aircraft metrics. *IEEE Trans. Eng. Manag.* 69 (6), 3864–3873.
- Kozanidis, G., Gavranis, A., Liberopoulos, G., 2014. Heuristics for flight and maintenance planning of mission aircraft. *Ann. Oper. Res.* 221 (1), 211–238.
- Kraus, M., Feuerriegel, S., 2019. Forecasting remaining useful life: Interpretable deep learning approach via variational Bayesian inferences. *Decis. Support Syst.* 125, 105551.
- Kurz, J., 2016. Capacity planning for a maintenance service provider with advanced information. *Eur. J. Oper. Res.* 251 (2), 466–477.
- Lagos, C., Delgado, F., Klapp, M.A., 2020. Dynamic optimization for airline maintenance operations. *Transp. Sci.* 54 (4), 998–1015.
- Levi, R., Magnanti, T., Muckstadt, J., Segev, D., Zarybnisky, E., 2014. Maintenance scheduling for modular systems: Modeling and algorithms. *Naval Research Logistics (NRL)* 61 (6), 472–488.
- Liang, Z., Feng, Y., Zhang, X., Wu, T., Chaovalitwongse, W.A., 2015. Robust weekly aircraft maintenance routing problem and the extension to the tail assignment problem. *Transp. Res. B Methodol.* 78, 238–259.
- Liang, Z., Xiao, F., Qian, X., Zhou, L., Jin, X., Lu, X., Karichery, S., 2018. A column generation-based heuristic for aircraft recovery problem with airport capacity constraints and maintenance flexibility. *Transp. Res. B Methodol.* 113, 70–90.
- Liu, M., Liang, B., Zheng, F., Chu, F., 2019. Stochastic airline fleet assignment with risk aversion. *IEEE Trans. Intell. Transp. Syst.* 20 (8), 3081–3090.
- Ma, H.L., Sun, Y., Chung, S.H., Chan, H.K., 2022. Tackling uncertainties in aircraft maintenance routing: a review of emerging technologies. *Transportation Research Part e: Logistics and Transportation Review* 164, 102805.

- Maher, S.J., Desaulniers, G., Soumis, F., 2018. The daily tail assignment problem under operational uncertainty using look-ahead maintenance constraints. *Eur. J. Oper. Res.* 264 (2), 534–547.
- Munari, P., Alvarez, A., 2019. Aircraft routing for on-demand air transportation with service upgrade and maintenance events: Compact model and case study. *J. Air Transp. Manag.* 75, 75–84.
- Nam, S., Song, W.-K., Yoon, H., 2023. An maintenance, repair, and overhaul (MRO) safety oversight system analysis: a case in Korea. *J. Air Transp. Manag.* 107, 102349.
- Niu, H., Zeng, J., Shi, H., Zhang, X., Liang, J., 2023. Degradation modeling and remaining useful life prediction for a multi-component system with stochastic dependence. *Comput. Ind. Eng.* 175, 108889.
- Öhman, M., Hiltunen, M., Virtanen, K., Holmström, J., 2021. Frontlog scheduling in aircraft line maintenance: from explorative solution design to theoretical insight into buffer management. *J. Oper. Manag.* 67 (2), 120–151.
- Özener, O.Ö., Örmeci Matoglu, M., Erdoğan, G., Haouari, M., Sözer, H., 2017. Solving a large-scale integrated fleet assignment and crew pairing problem. *Ann. Oper. Res.* 253 (1), 477–500.
- Patriarca, R., Lovaglio, L., Simone, F., 2025. Functional resonance analysis via a genetic algorithm to ensure cost-effective maintenance planning. *Int. J. Prod. Econ.* 281, 109516.
- Patriksson, M., Strömberg, A.-B., Wojciechowski, A., 2015. The stochastic opportunistic replacement problem, part II: a two-stage solution approach. *Ann. Oper. Res.* 224 (1), 51–75.
- Qin, Y., Ng, K.K.H., 2023. Analysing the impact of collaborations between airlines and maintenance service company under MRO outsourcing mode: Perspective from airline's operations. *J. Air Transp. Manag.* 109, 102396.
- Qin, Y., Chan, F.T.S., Chung, S.H., Qu, T., Niu, B., 2018. Aircraft parking stand allocation problem with safety consideration for independent hangar maintenance service providers. *Comput. Oper. Res.* 91, 225–236.
- Qin, Y., Zhang, J.H., Chan, F.T.S., Chung, S.H., Niu, B., Qu, T., 2020. A two-stage optimization approach for aircraft hangar maintenance planning and staff assignment problems under MRO outsourcing mode. *Comput. Ind. Eng.* 146, 106607.
- Qin, Y., Ng, K.K.H., Sun, X., 2024. Service accessibility strategy for aircraft maintenance routing with interorganizational collaborations on outsourcing maintenance. *Comput. Ind. Eng.* 190, 110071.
- Regattieri, A., Giazzi, A., Gamberi, M., Gamberini, R., 2015. An innovative method to optimize the maintenance policies in an aircraft: General framework and case study. *J. Air Transp. Manag.* 44–45, 8–20.
- Ruther, S., Boland, N., Engineer, F.G., Evans, I., 2017. Integrated aircraft routing, crew pairing, and tail assignment: Branch-and-price with many pricing problems. *Transp. Sci.* 51 (1), 177–195.
- Safaei, N., Jardine, A.K.S., 2018. Aircraft routing with generalized maintenance constraints. *Omega* 80, 111–122.
- Saltzman, R.M., Stern, H.I., 2022. The multi-day aircraft maintenance routing problem. *J. Air Transp. Manag.* 102, 102224.
- Sanchez, D.T., Boyaci, B., Zografos, K.G., 2020. An optimisation framework for airline fleet maintenance scheduling with tail assignment considerations. *Transportation Research Part B: Methodological* 133, 142–164.
- Savitha, R., Ambikapathi, A., Rajaraman, K., 2020. Online RBM: growing restricted boltzmann machine on the fly for unsupervised representation. *Appl. Soft Comput.* 92, 106278.
- Sciau, J.-B., Goyon, A., Sarazin, A., Bascans, J., Prud'homme, C., & Lorca, X. (2024). Using constraint programming to address the operational aircraft line maintenance scheduling problem. *Journal of Air Transport Management*, 115, 102537.
- Seif, J., Yu, A.J., 2018. An extensive operations and maintenance planning problem with an efficient solution method. *Comput. Oper. Res.* 95, 151–162.
- Shao, S., Sherali, H.D., Haouari, M., 2017. A novel model and decomposition approach for the integrated airline fleet assignment, aircraft routing, and crew pairing problem. *Transp. Sci.* 51 (1), 233–249.
- Sharma, G., Rai, R.N., 2021. Failure modes based censored data analysis for repairable systems and its industrial perspective. *Comput. Ind. Eng.* 158, 107439.
- Shaukat, S., Katscher, M., Wu, C.-L., Delgado, F., Larrain, H., 2020. Aircraft line maintenance scheduling and optimisation. *J. Air Transp. Manag.* 89, 101914.
- Singh, V.P., Ganguly, K.K., Hussain, S.M.F., 2024. Understanding no fault found event risk in military aircraft MRO planning: a step towards robust bidding and contract finalization. *Int. J. Prod. Econ.* 275, 109344.
- Sun, X., Zhao, X., Chung, S.H., Ma, H.L., 2025. An interactive decision making framework design for the outsourcing cooperation between the service provider and the airline: an exact bilevel method. *Transportation Research Part e: Logistics and Transportation Review* 199, 104148.
- Van Kessel, P.J., Freeman, F.C., Santos, B.F., 2023. Airline maintenance task rescheduling in a disruptive environment. *Eur. J. Oper. Res.* 308 (2), 605–621.
- Varena, S., Li, H., Ribeiro, M., Santos, B.F., 2025. Stochastic discrete event simulation of airline network and maintenance operations. *J. Air Transp. Manag.* 125, 102789.
- Villafranca, M., Delgado, F., Klapp, M., 2025. Aircraft maintenance scheduling under uncertain task processing time. *Transportation Research Part e: Logistics and Transportation Review* 196, 104012.
- Vink, J., Santos, B.F., Verhagen, W.J.C., Medeiros, I., Filho, R., 2020. Dynamic aircraft recovery problem—An operational decision support framework. *Comput. Oper. Res.* 117, 104892.
- Wang, L., Li, B., Zhao, X., 2024. Multi-objective predictive maintenance scheduling models integrating remaining useful life prediction and maintenance decisions. *Comput. Ind. Eng.* 197, 110581. <https://doi.org/10.1016/j.cie.2024.110581>.
- Weide, T.V.D., Deng, Q., Santos, B.F., 2022. Robust long-term aircraft heavy maintenance check scheduling optimization under uncertainty. *Comput. Oper. Res.* 141, 105667.
- Witteaman, M., Deng, Q., Santos, B.F., 2021. A bin packing approach to solve the aircraft maintenance task allocation problem. *Eur. J. Oper. Res.* 294 (1), 365–376.
- Xu, A., Wang, R., Weng, X., Wu, Q., Zhuang, L., 2025. Strategic integration of adaptive sampling and ensemble techniques in federated learning for aircraft engine remaining useful life prediction. *Appl. Soft Comput.* 175, 113067.
- Yang, D., Wang, H., Feng, Q., Ren, Y., Sun, B., Wang, Z., 2018. Fleet-level selective maintenance problem under a phased mission scheme with short breaks: a heuristic sequential game approach. *Comput. Ind. Eng.* 119, 404–415.
- Yang, H.J., Jeong, S.J., Yoon, S.W., 2020. Enhancement for human resource management in the ULD build-up process of air-cargo terminal: a strategic linkage approach. *J. Heuristics* 26 (2), 301–333.
- Zhang, Q., Yao, W., Xu, P., Fang, Z., 2023. Optimal age replacement policies of mission-oriented systems with discounting. *Comput. Ind. Eng.* 177, 109027.
- Zhang, Q., Chung, S.-H., Ma, H.-L., Sun, X., 2024. Robust aircraft maintenance routing with Heterogeneous aircraft maintenance tasks. *Transp. Res. Part C Emerging Technol.* 160, 104518.
- Zhang, M., Yang, J., Zhang, C., He, S., Liu, H., Wang, J., Wang, Z., 2025. An approximate dynamic programming approach for solving aircraft fleet engine maintenance problem: Methodology and a case study. *Eur. J. Oper. Res.* 321 (3), 958–973.